Magnetic Fields

A compass points north. So why is the Earth's north a magnetic south pole?



Advanced Higher Physics

PART 1

Where magnetism comes from, and how to sketch it

Every magnet is made of moving electrons.

By the end of this part…

1. I can **explain ferromagnetism** in the spec's words.
2. I can **sketch** the field between poles, round a **solenoid** and round Earth.
3. I can **compare** the four forces: strength and range.

From Higher — moving charge makes a field

Moving charge

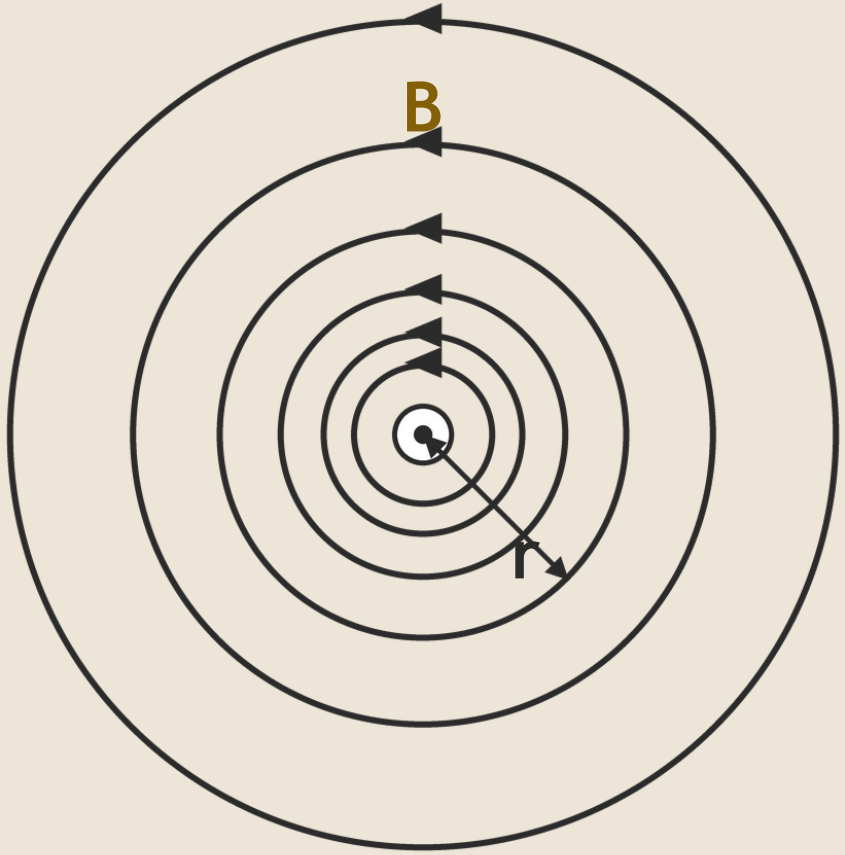
Makes a magnetic field.

A wire

The field circles the current.

New at AH

How strong: $B = \mu_0 I / 2\pi r$.



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Every electron is a tiny magnet

Electrons

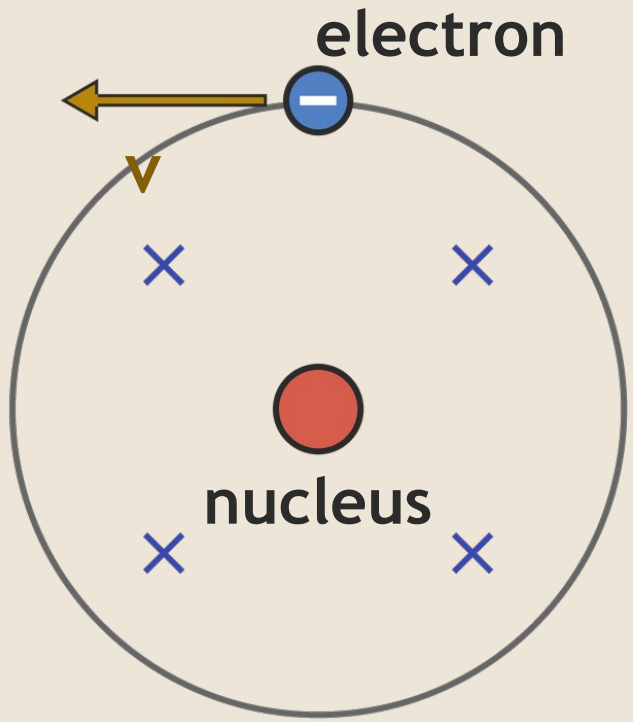
Move round the nucleus.

Moving charge

So each one has a magnetic effect.

Most materials

The effects point every way and cancel.



B into page

Ferromagnetism — domains line up

Iron, nickel, cobalt

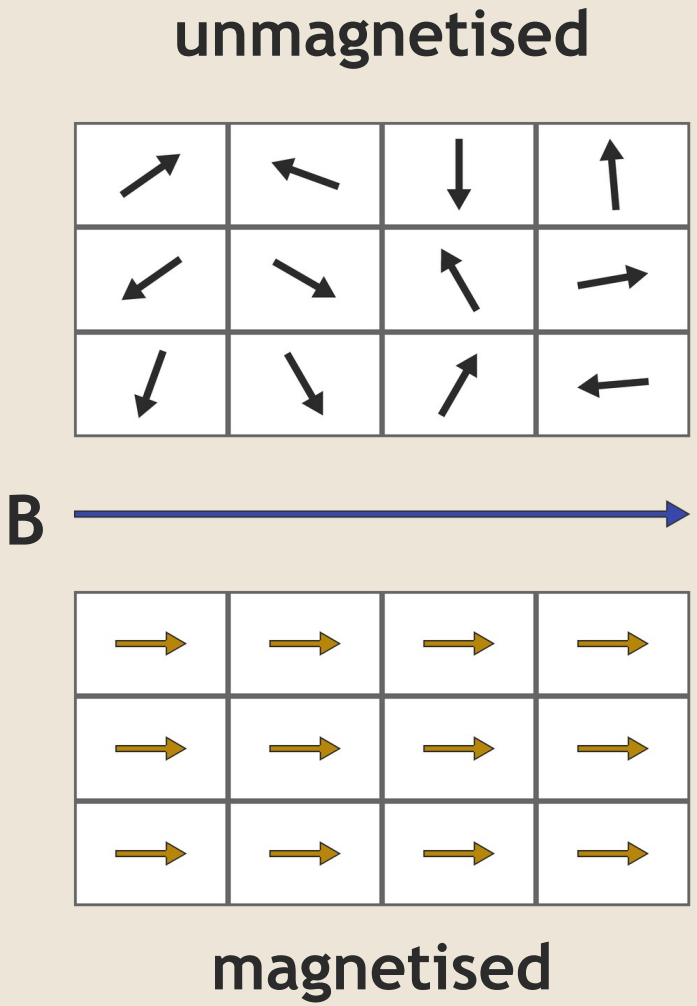
and some rare earths.

Domains

Regions of dipoles lined up.

Outside field

Lines the domains up: magnetised.



Ferromagnetism

LEARN THIS WORD FOR WORD

Iron, nickel, cobalt and some rare earths show **ferromagnetism**: their **magnetic dipoles can be made to align**, so the material becomes **magnetised**.

Field lines — the rules

Direction

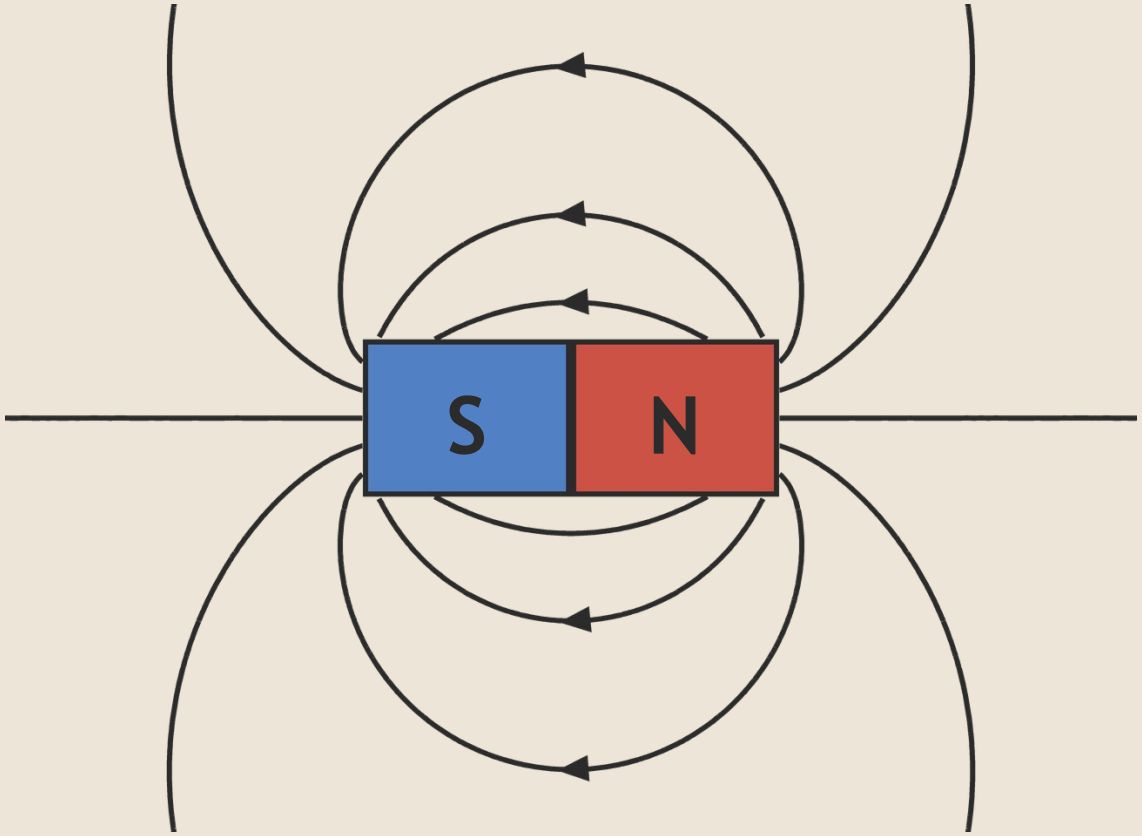
Out of N, into S.

Lines

Never cross.

Spacing

Closer lines: stronger field.



Between two magnets

Unlike poles

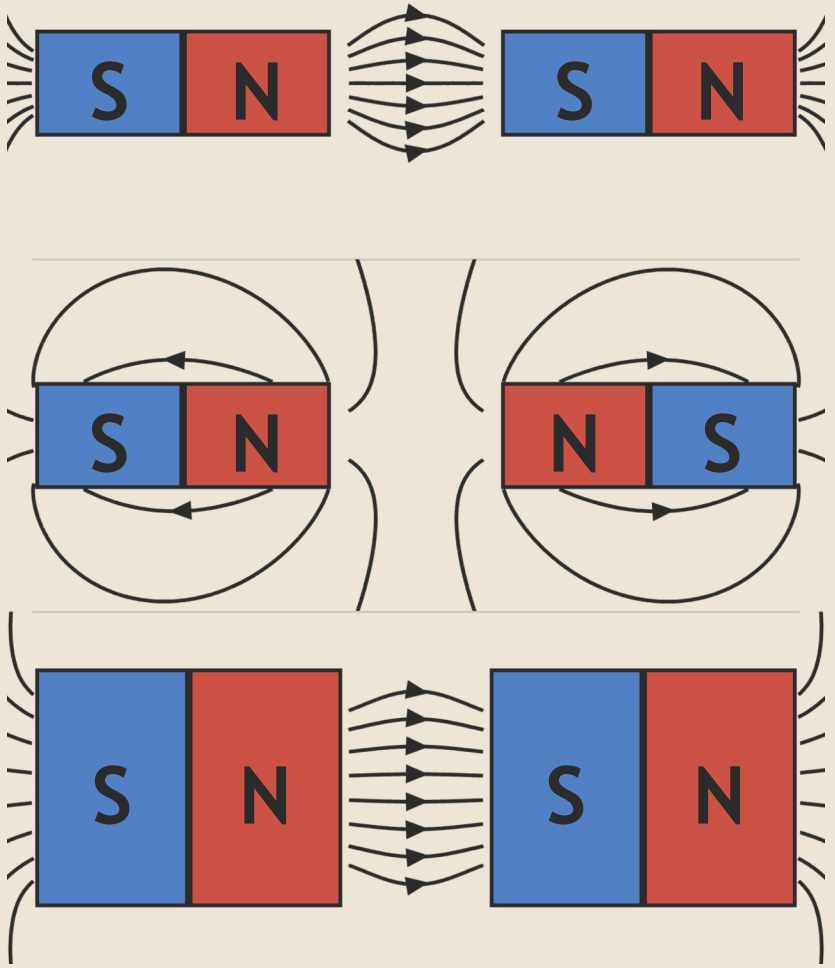
Lines cross the gap, N to S.

Like poles

Lines push apart.

Flat poles

Straight, parallel lines: uniform.



Round a solenoid

Inside

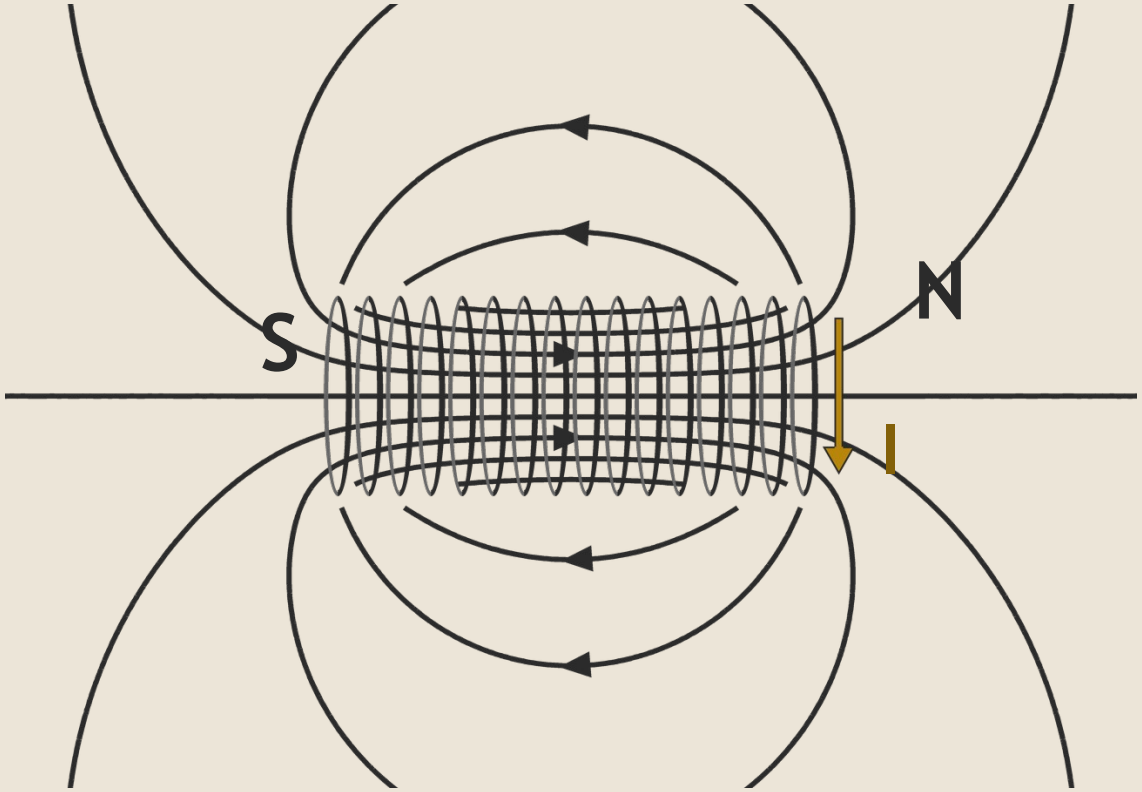
Straight, parallel, evenly spaced.

Outside

Like a bar magnet.

N end

Right hand: fingers with I, thumb to N.



An iron core

Same coil

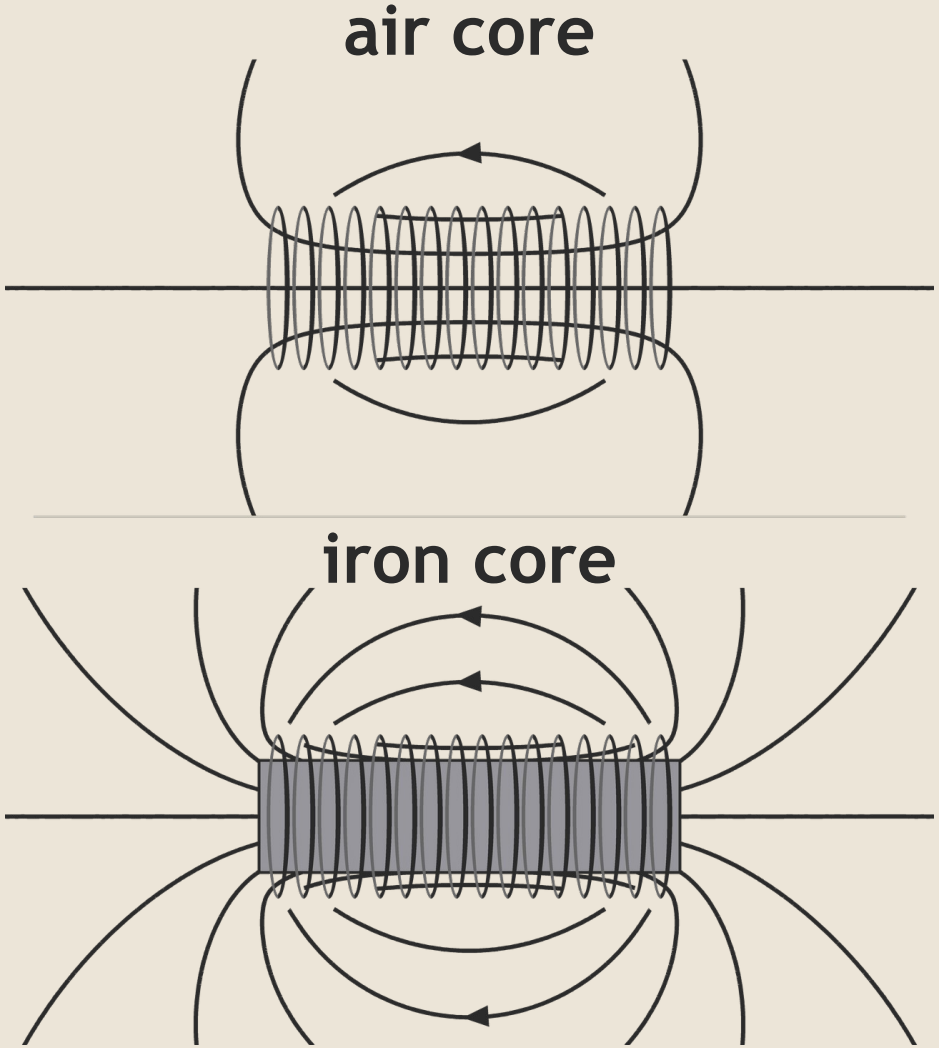
Same current.

Iron core

Domains line up and add their field.

Field lines

More densely packed.



Round the Earth

Like a bar magnet

Field lines leave near the South Pole.

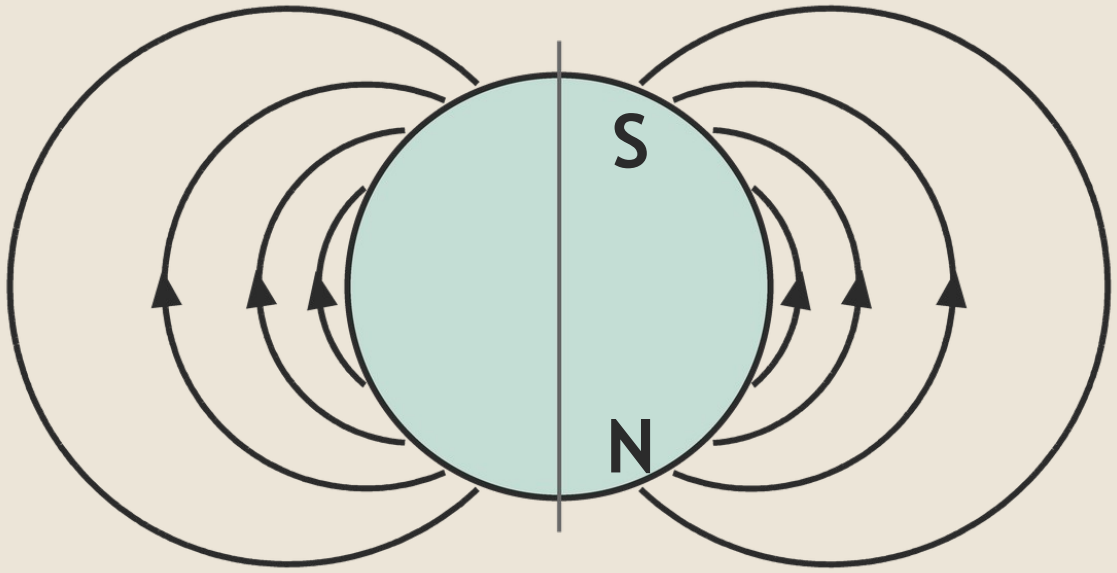
Geographic North

Is a magnetic **south** pole.

A compass

Its N end is pulled there.

geographic North Pole



geographic South Pole

Exam question — 2024 Q14(a)

14. A solenoid is a helical coil of wire. When there is a current in the wire a magnetic field is produced in and around the solenoid.

(a) **Figure 14A** shows a solenoid.

On **Figure 14A**, sketch the magnetic field pattern produced by the current. **2**

(An additional diagram, if required, can be found on *page 53*.)

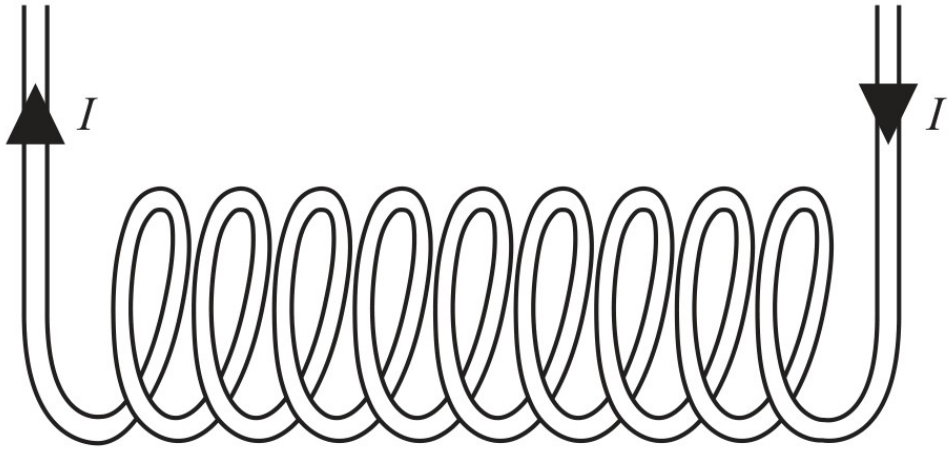
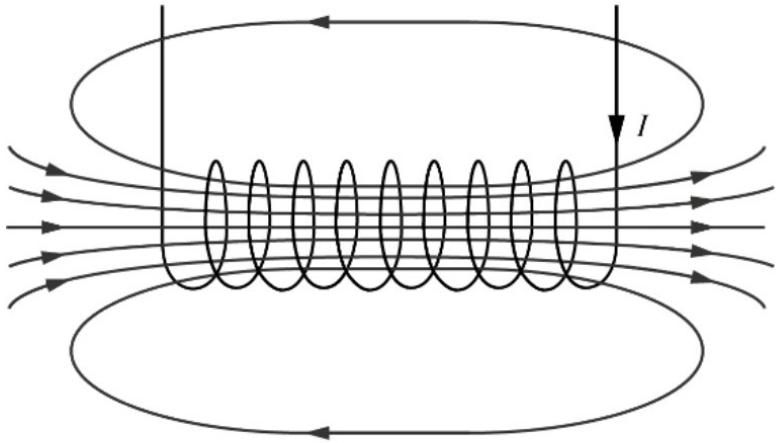


Figure 14A

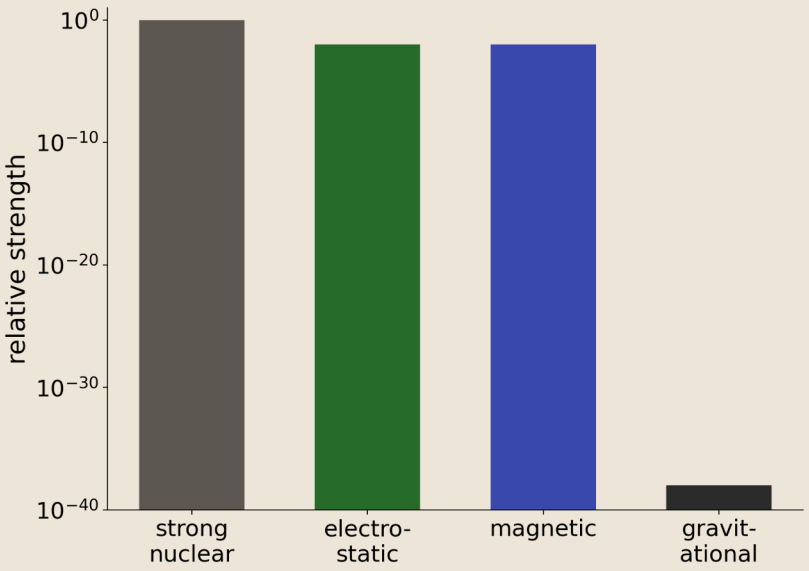
How it's marked — 2024 Q14(a)

Question			Expected response	Max mark	Additional guidance
14.	(a)			2	1 mark for shape 1 mark for direction

Four forces compared

Force	Strength	Range
Strong nuclear	1	$\approx 10^{-15}$ m
Electrostatic	$\approx 10^{-2}$	infinite
Magnetic	$\approx 10^{-2}$	infinite
Gravitational	$\approx 10^{-38}$	infinite

Electrostatic and magnetic: one force, electromagnetism.



Exam question — 2024 Q14(b) (1 of 2)

(b) For a short, wide solenoid, the magnetic induction produced depends on the current I in the solenoid, the radius R of the solenoid, the number of turns N of the coil, and the distance x along the central axis of the solenoid from the centre of the solenoid.

This solenoid is shown in **Figure 14B**.

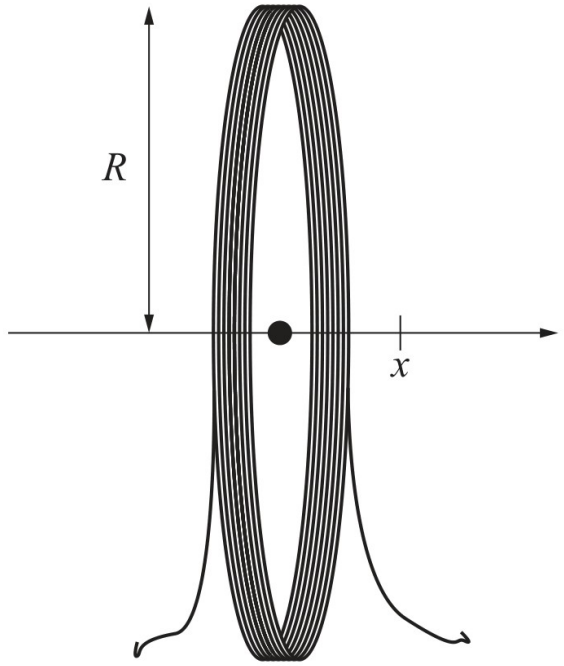


Figure 14B

Exam question — 2024 Q14(b) (2 of 2)

The magnetic induction of this type of solenoid is given by the relationship

$$B = \frac{\mu_0 INR^2}{2\left(R^2 + x^2\right)^{\frac{3}{2}}}$$

There is a current of 3.5 A in the wire.

The radius of the solenoid is 120 mm.

The solenoid has 64 turns.

The maximum magnetic induction occurs at the centre of the solenoid, where $x = 0$ m.

Determine the maximum magnetic induction produced by this solenoid. **2**

Space for working and answer

Marking — 2024 Q14(b)

Question			Expected response	Max mark	Additional guidance
	(b)		$B = \dfrac{\mu_0 INR^2}{2\left(R^2 + x^2\right)^{\frac{3}{2}}}$	2	Accept: 1, 1.17, 1.173
			$B = \dfrac{4\pi \times 10^{-7} \times 3.5 \times 64 \times (120 \times 10^{-3})^2}{2\left((120 \times 10^{-3})^2 + 0^2\right)^{\frac{3}{2}}}$ **(1)**		Substitution of x = 0 not required for first mark
			$B = 1.2 \times 10^{-3}$ T **(1)**		

Exam question — 2024 Q14(c)

(c) A ferromagnetic material can be inserted into a solenoid, along its central axis, to increase the magnetic induction.

(i) State what is meant by the term *ferromagnetic*. **1**

(ii) Suggest a ferromagnetic material that will increase the magnetic induction. **1**

(iii) Predict the effect that the insertion of this ferromagnetic material will have on the magnetic field lines around the solenoid. **1**

Marking — 2024 Q14(c)

Question			Expected response	Max mark	Additional guidance
	(c)	(i)	(In a ferromagnetic material) the magnetic dipoles/domains (in the material) can be made to align, (resulting in the material becoming magnetised).	**1**	
		(ii)	Iron (or nickel or cobalt)	**1**	Any other ferromagnetic material is acceptable, such as steel or neodymium
		(iii)	The field lines will be more densely/ tightly packed together **OR** there will be more of them.	**1**	

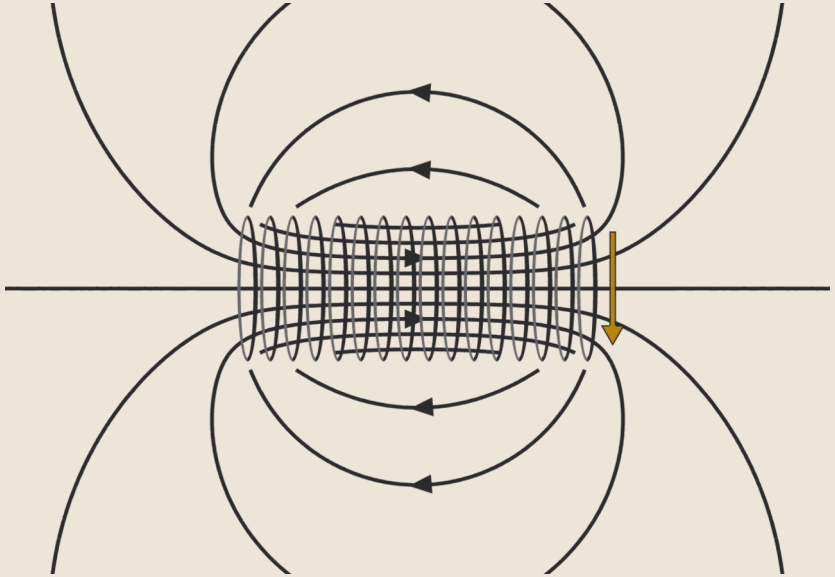
What Candidates Get Wrong

Inside the coil: straight, parallel lines (2024 Q14(a) report).

Arrows: a mark of their own (2024 Q14(a) MI).

Ferromagnetic: dipoles **can be made to align** (2024 Q14(c)(i): few could).

Iron core: lines **more densely packed** (2024 Q14(c)(iii) MI).



Look back — by the end of this part…



1. I can **explain ferromagnetism** in the spec's words.
2. I can **sketch** the field between poles, round a **solenoid** and round Earth.
3. I can **compare** the four forces: strength and range.

PART 2

The field round a wire, and the force on a wire

Two cables under the sea pull on each other. How hard?

By the end of this part…

1. I can use $B = \mu_0 I / 2\pi r$.
2. I can use $F = IlB \sin \theta$.
3. I can **calculate** the force between two wires.
4. I can **explain** when there is **no force**.

Magnetic induction near a wire

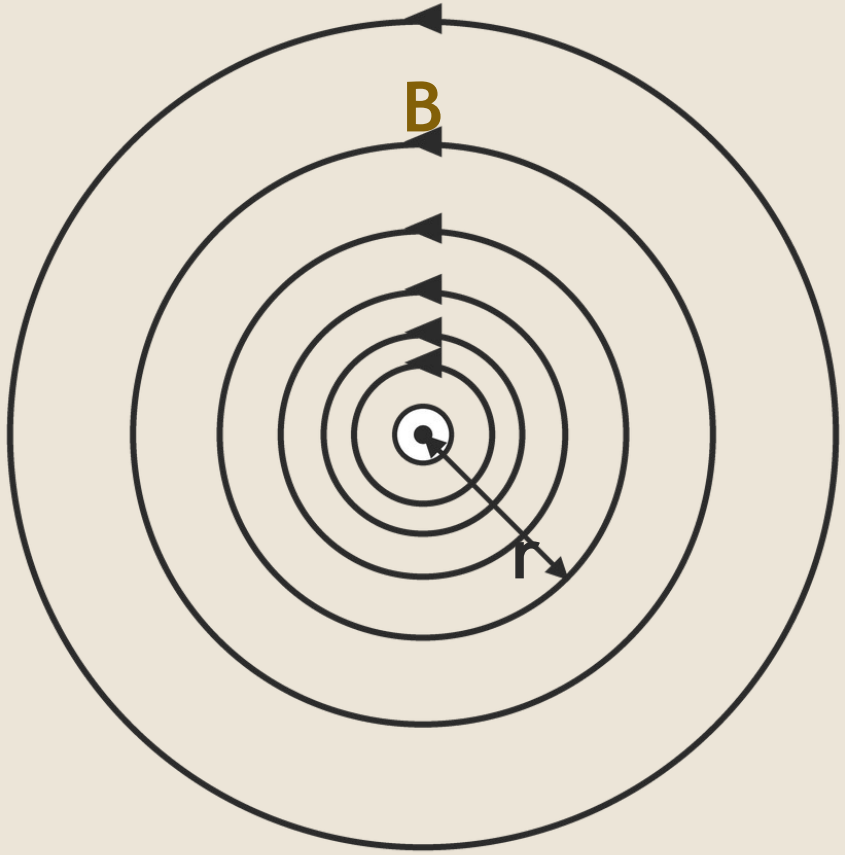
$$B = \mu_0 I / 2\pi r$$

B: magnetic induction (T)

I: current (A)

r: distance from the wire (m)

$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} (data sheet)



centre: I out of page

B falls as 1/r

Double r

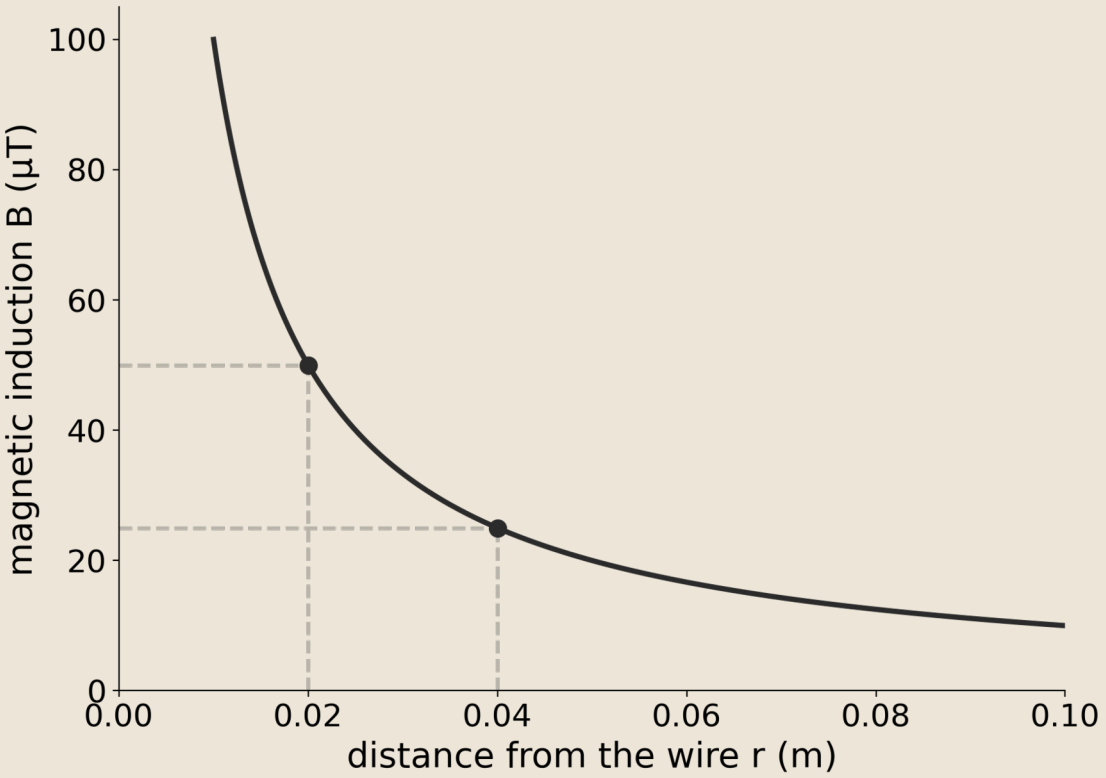
Halve B.

Close to the wire

Strongest field.

Field lines

Spread out as r grows.



I = 5.0 A

Worked example 1 — B near a cable

A long straight cable carries a current of **250 A**.

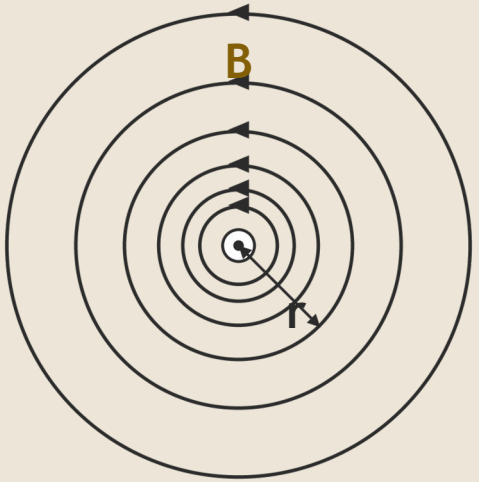
Calculate the magnetic induction **0.40 m** from the cable. **3**

$$B = \mu_0 I / 2\pi r$$

$$B = 4\pi \times 10^{-7} \times 250 / (2\pi \times 0.40)$$

$$B = 1.3 \times 10^{-4}\ T$$

3 marks: relationship, substitution, answer with unit.



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Exam question — 2022 Q15(a)(i) (1 of 2)

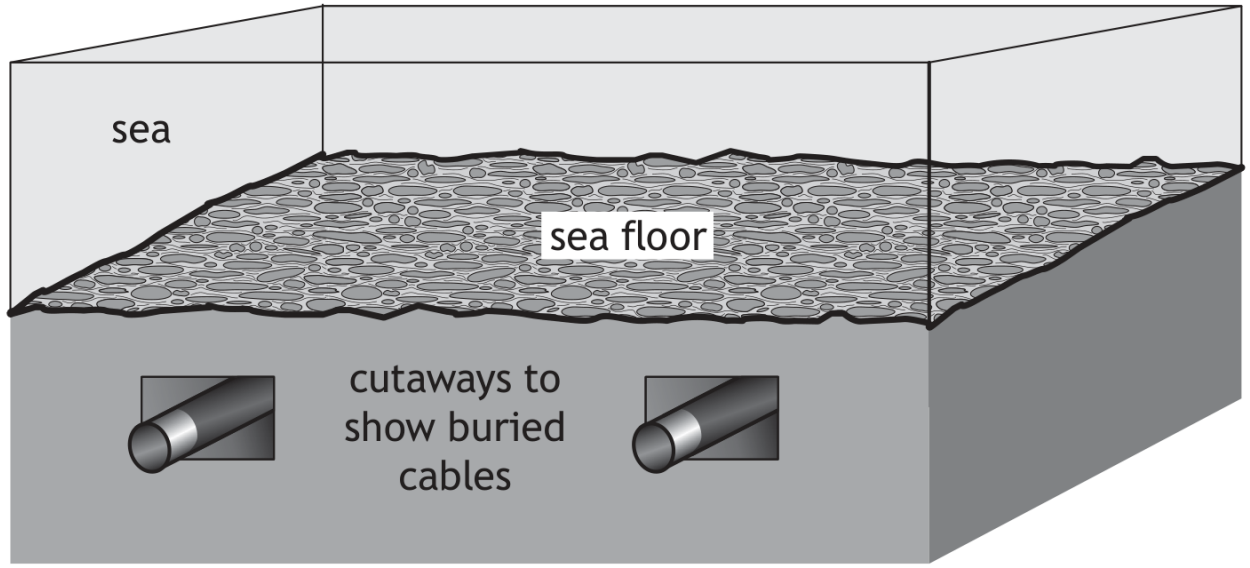
15. An undersea high voltage DC electrical power link consists of two cables buried under the seabed.

The magnetic permeability of the seabed can be taken to be the same as the permeability of free space.

There is a current of 1.80 kA in each cable.

The cables are buried 30.0 m apart, as shown in Figure 15A.

not to scale



Exam question — 2022 Q15(a)(i) (2 of 2)

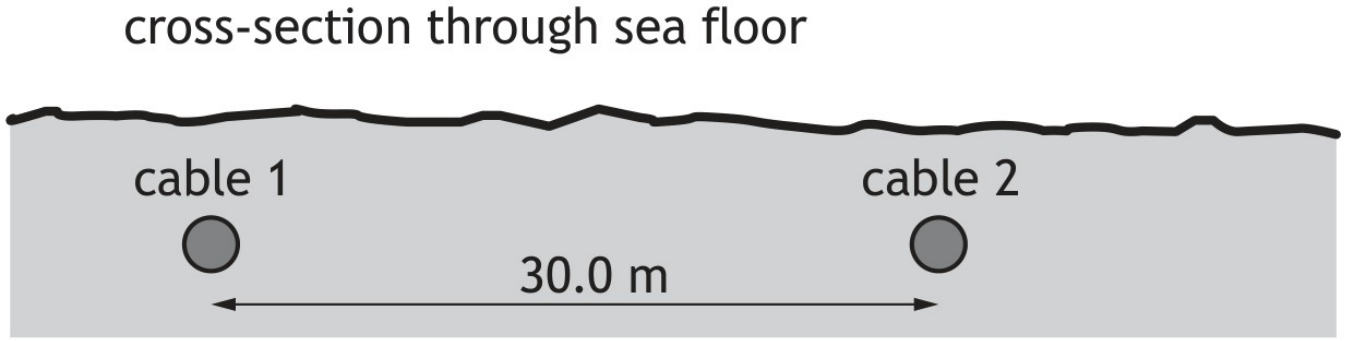


Figure 15A

(a) (i) Calculate the magnetic induction at cable 2 due to the current in cable 1. **3**

Space for working and answer

How it's marked — 2022 Q15(a)(i)

Question			Expected response		Max mark	Additional guidance
15.	(a)	(i)	$B = \frac{\mu_0 I}{2\pi r}$	(1)	3	Accept: 1, 1.20, 1.200
			$B = \frac{4\pi \times 10^{-7} \times 1.8 \times 10^{3}}{2\pi \times 30.0}$	(1)		
			$B = 1.2 \times 10^{-5}$ T	(1)		

Force on a wire in a field

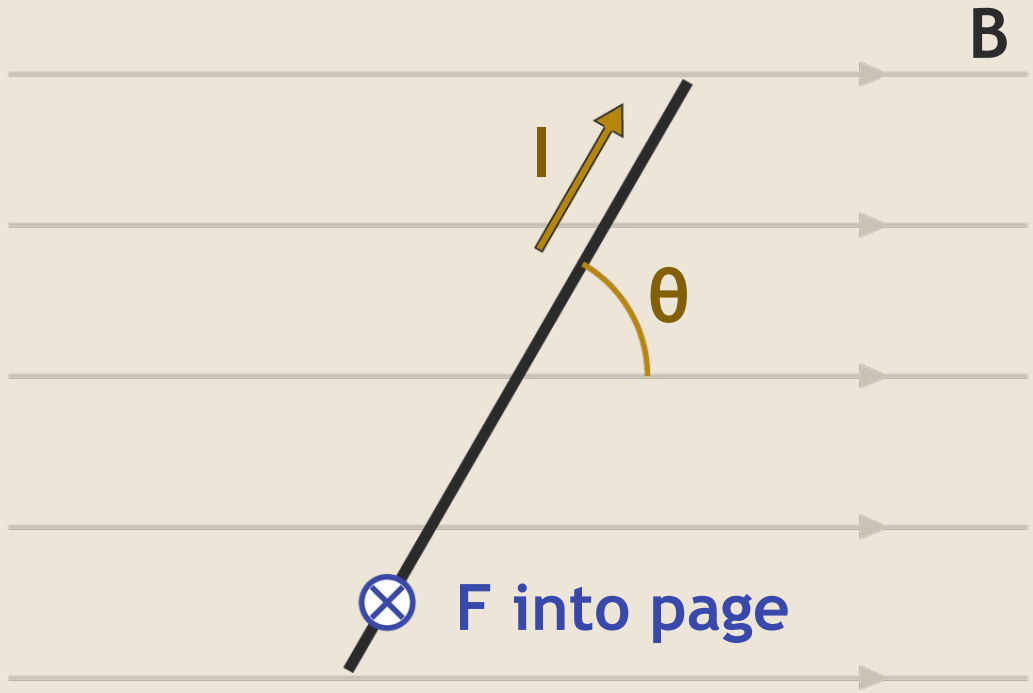
$$F = IlB \sin \theta$$

F: force (N)

I: current (A), l: length (m)

B: magnetic induction (T)

θ: angle between wire and B



What 1 tesla means

1 T

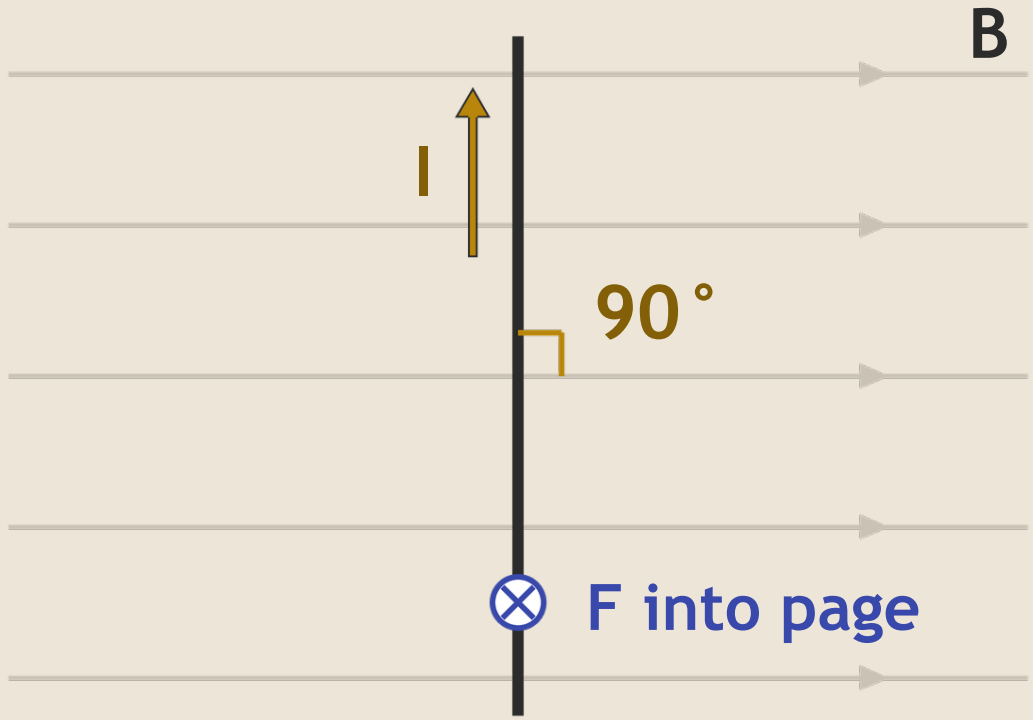
1 m of wire, 1 A, at 90° to B: a force of 1 N.

At 90°

$\sin\theta = 1$, so $F = IlB$.

Typical

A school magnet: about 0.1 T.



Which way is the force?

At 90°

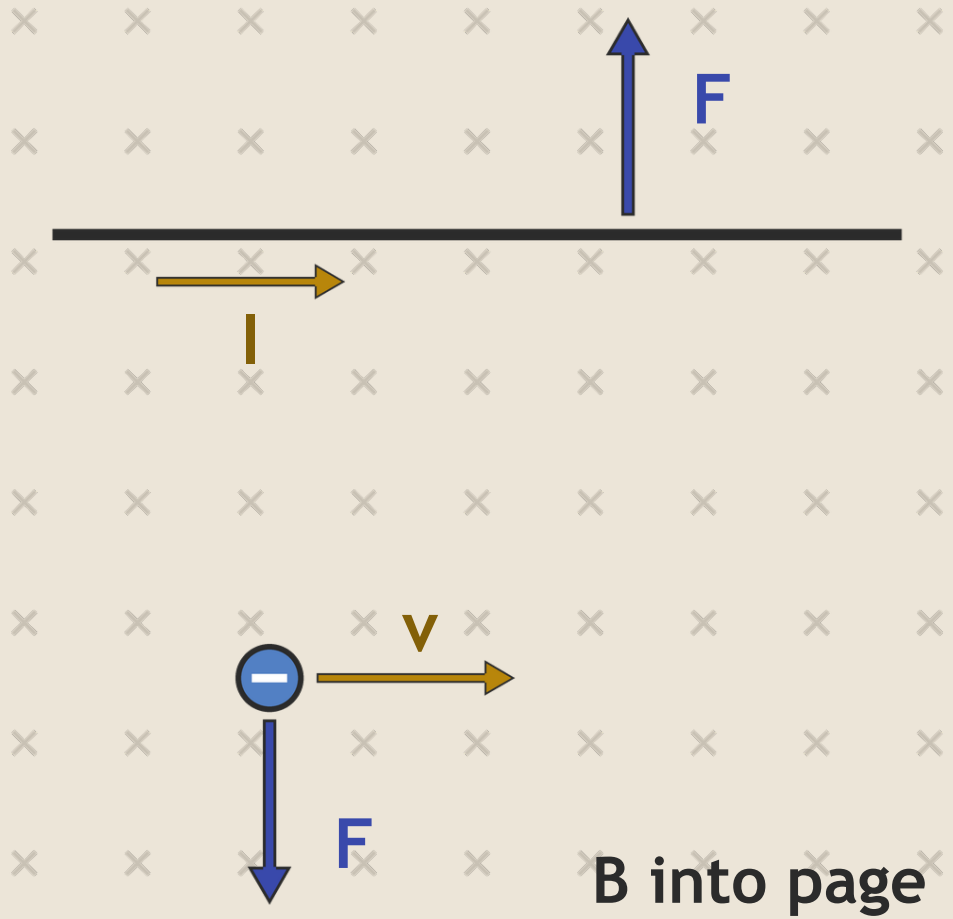
F is at right angles to I and to B.

Negative charge

Right-hand rule (Higher).

Current I or + charge

Force the **opposite** way.



No force when the wire is along B

θ = 0

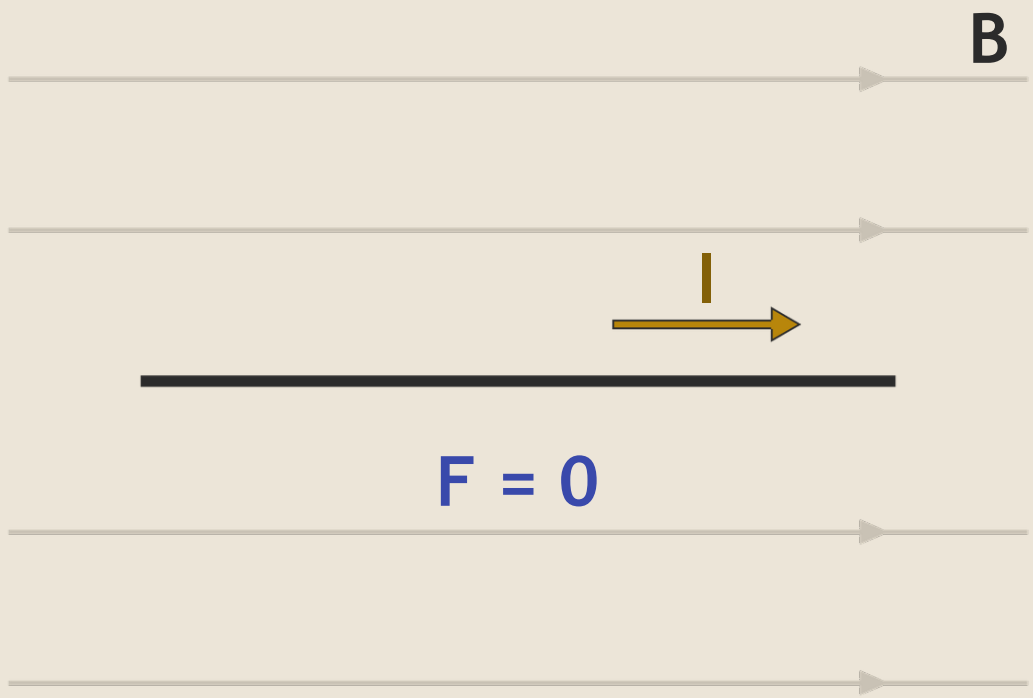
sin 0 = 0, so F = 0.

Current

Parallel to the field.

Motor side QR

No force (2026 Q15).



Your lab — F = BIL with a balance

Force

F = Δm g from the balance.

Graph

F against I: a straight line.

Gradient

Gradient = Bl, so B = gradient / l.

The lab page

Worked example 2 — a wire at an angle

A wire **50 mm** long carries **3.0 A**. It is at **30°** to a magnetic field of **0.20 T**.

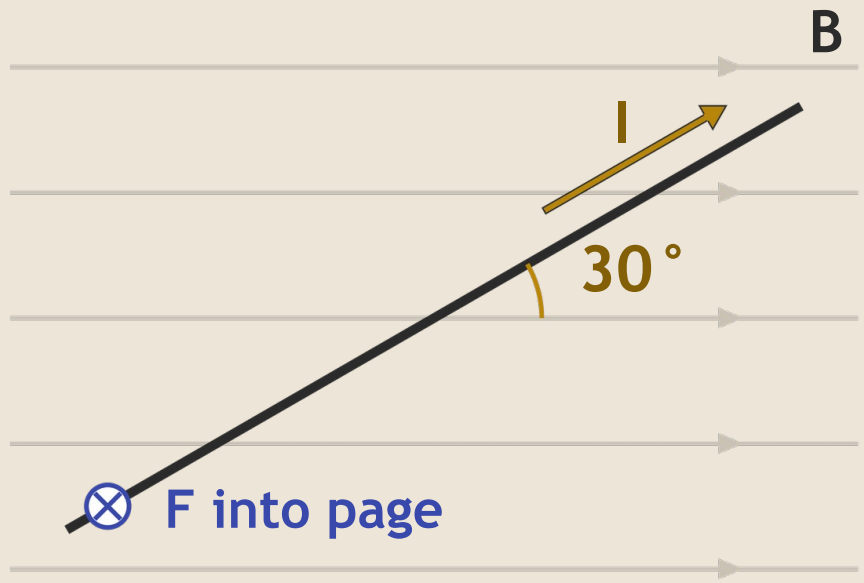
Calculate the force on the wire. **3**

$$F = IlB \sin \theta$$

$$F = 3.0 \times 0.050 \times 0.20 \times \sin 30^\circ$$

$$F = 1.5 \times 10^{-2} \text{ N}$$

3 marks: relationship, substitution, answer with unit.



Two parallel wires

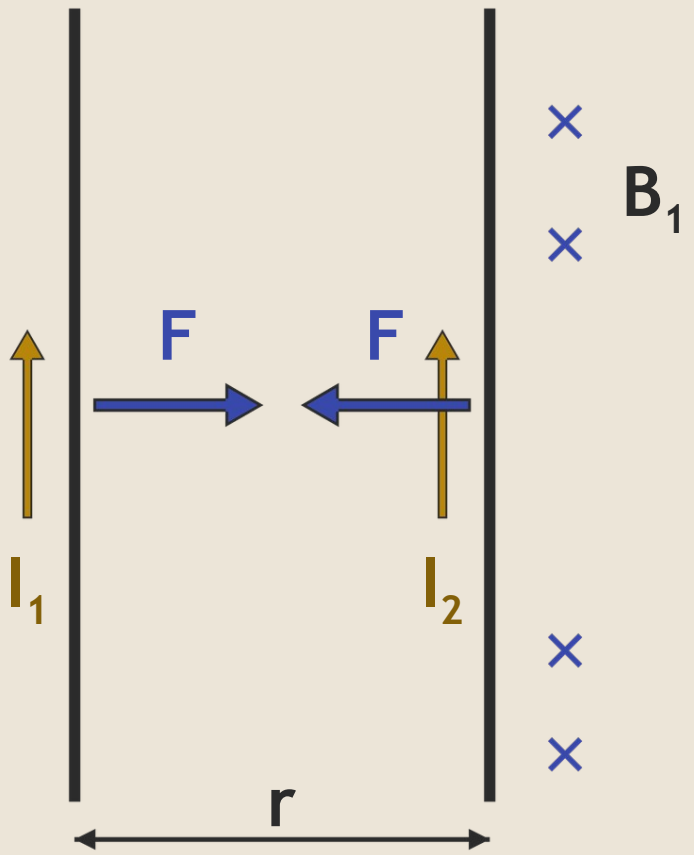
$$B = \mu_0 I / 2\pi r$$

$$F/l = I_2 B_1$$

B_1: field of wire 1 at wire 2

Same direction: attract.

Opposite: repel.



Worked example 3 — force between cables

Two long parallel cables are **0.25 m** apart. Each carries **400 A** in the same direction.

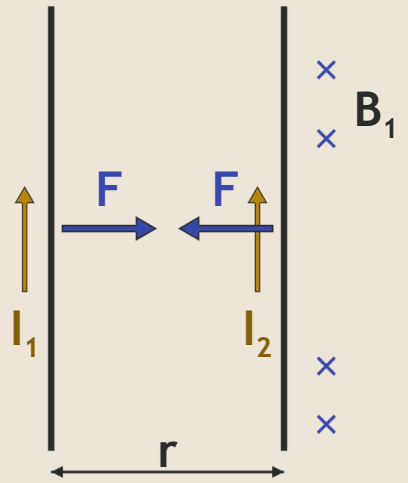
(a) **Calculate** the magnetic induction at cable 2 due to cable 1. **3**

$$B = \mu_0 I / 2\pi r$$

$$B = 4\pi \times 10^{-7} \times 400 / (2\pi \times 0.25)$$

$$B = 3.2 \times 10^{-4}\ T$$

3 marks: relationship, substitution, answer with unit.



Worked example 3 — continued

Two long parallel cables are **0.25 m** apart. Each carries **400 A** in the same direction.

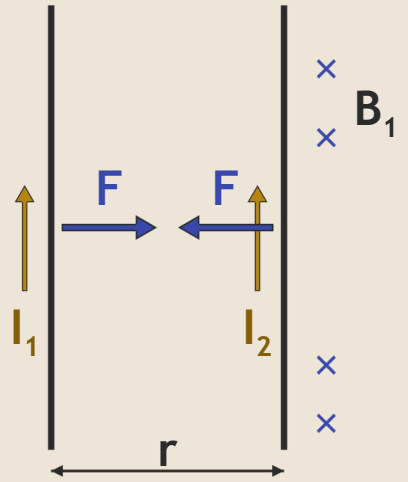
(b) **Determine** the force per unit length on cable 2. **3**

$$F/l = I_2 B_1$$

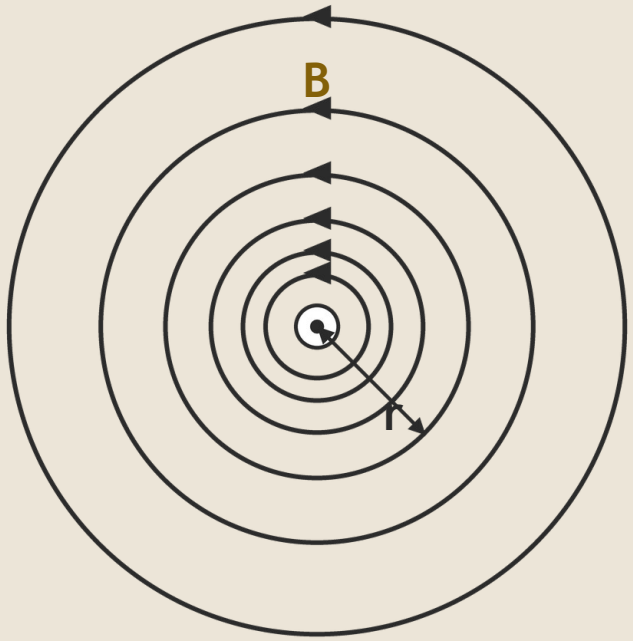
$$F/l = 400 \times 3.2 \times 10^{-4}$$

$$F/l = 0.13 \text{ N m}^{-1}, \text{ towards cable 1}$$

3 marks: relationship, substitution, answer with unit.



A wire carries **12 A**. **Calculate** the magnetic induction **30 mm** from the wire.



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$B = 8.0 \times 10^{-5}\ T$

Exam question — 2023 Q14(a) (1 of 2)

14. A student designs an experiment to determine the permeability of free space (μ_0) by measuring the magnetic induction around a current carrying wire.

The experimental setup is shown in **Figure 14A**.

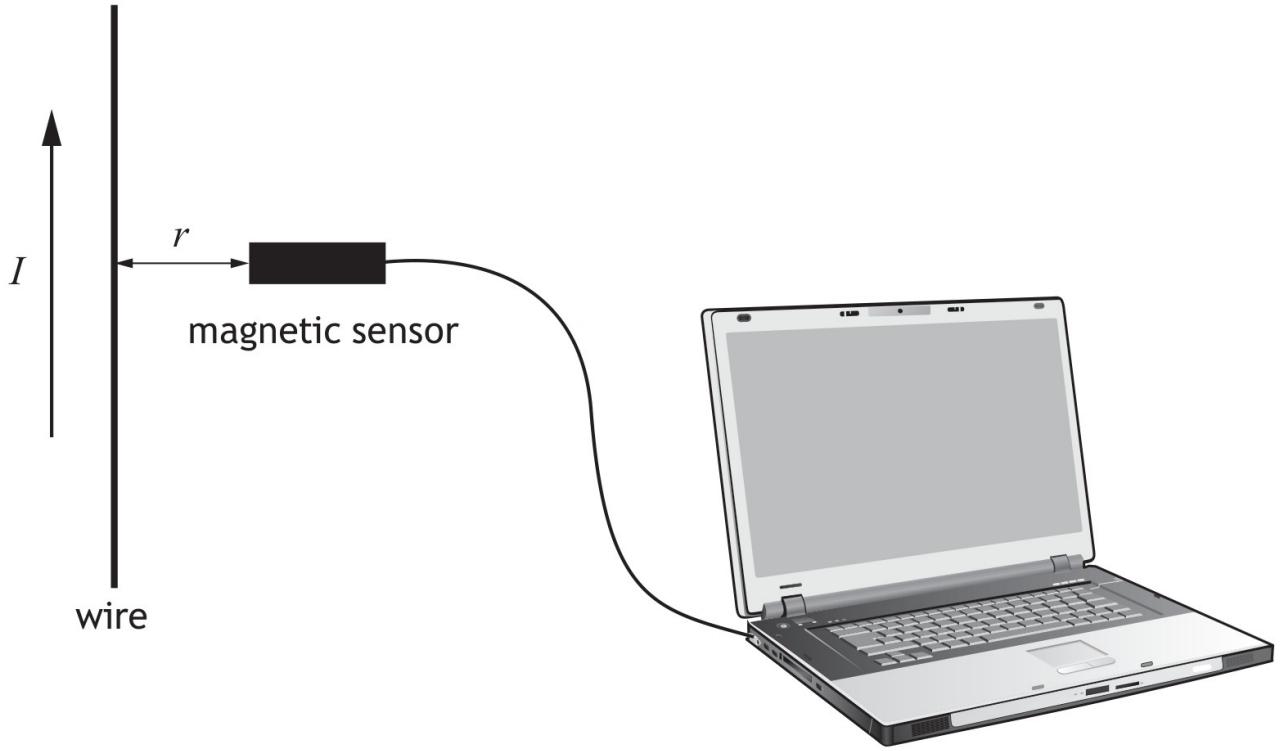


Figure 14A

Exam question — 2023 Q14(a) (2 of 2)

The front of the magnetic sensor is positioned 0.010 m from the wire.

(a) The student tests the setup using a current of 1.2 A in the wire. The student measures the magnetic induction due to the current in the wire to be 27 μT.

Show that the value obtained for μ_0 is 1.4×10^{-6} H m^{-1}. **2**

Space for working and answer

Marking — 2023 Q14(a)

Question			Expected response		Max mark	Additional guidance
14.	(a)		$B = \frac{\mu_0 I}{2\pi r}$	1	2	SHOW QUESTION
			$27 \times 10^{-6} = \frac{\mu_0 \times 1.2}{2\pi \times 0.010}$	1		
			$\mu_0 = 1.4 \times 10^{-6}\ \text{Hm}^{-1}$			

Exam question — 2026 Q15(a)(i)

15. A model electric motor consists of a rotor, which is free to rotate about an axis perpendicular to a uniform magnetic field, as shown in **Figure 15A.**

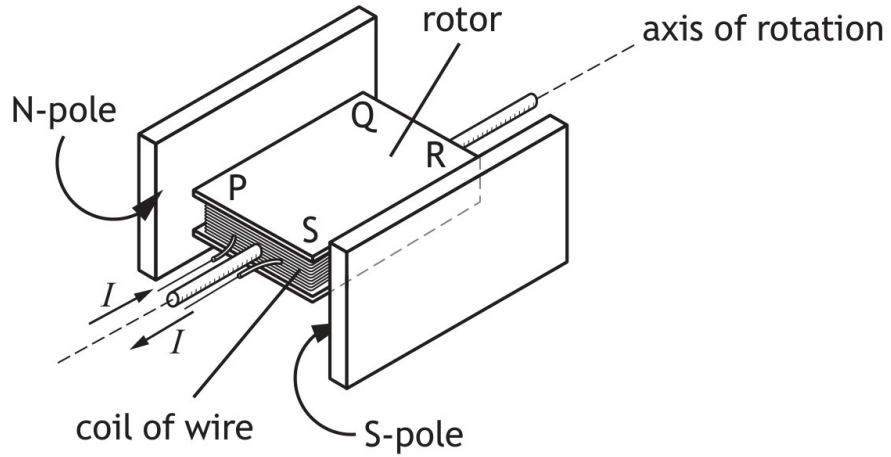


Figure 15A

The length of the wire along the side PQ is 45 mm.

The magnetic induction between the poles of the magnets is 44 mT.

The current in the coil of wire is 0.56 A.

(a) (i) Calculate the magnitude of the force experienced by one wire of the coil on the rotor side PQ. **3**

Space for working and answer

Marking — 2026 Q15(a)(i)

Question			Expected response		Max mark	Additional guidance
15.	(a)	(i)	$F = IlB(\sin\theta)$	**(1)**	3	Accept: 1, 1.11, 1.109
			$F = 0.56 \times 45 \times 10^{-3} \times 44 \times 10^{-3}$	**(1)**		
			$F = 1.1 \times 10^{-3}$ N	**(1)**		

Exam question — 2022 Q15(a) (1 of 2)

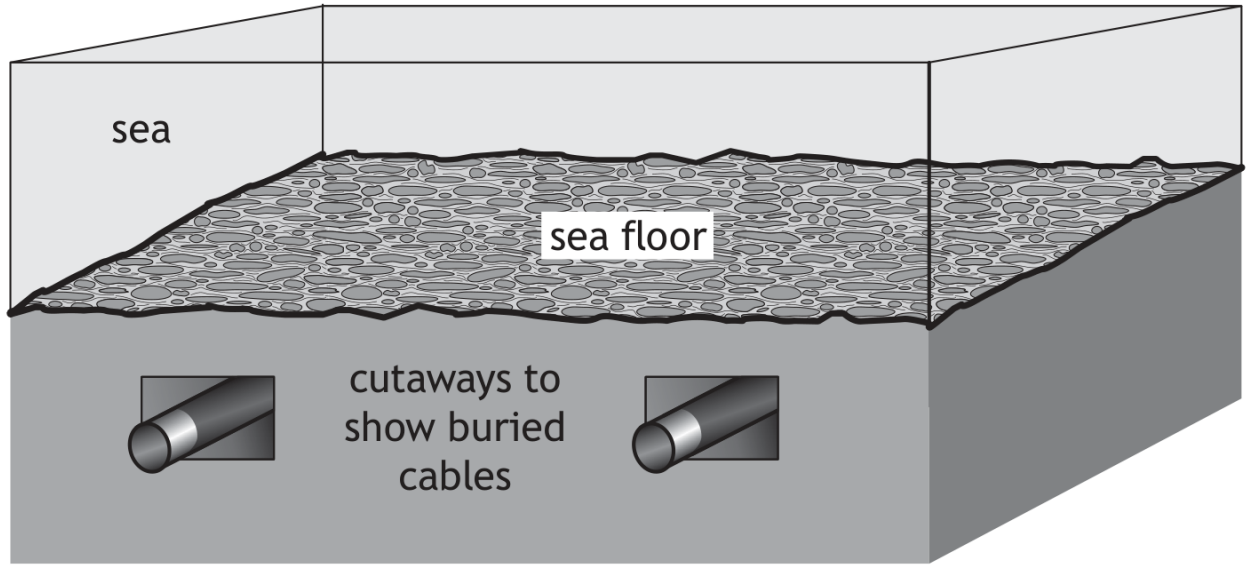
15. An undersea high voltage DC electrical power link consists of two cables buried under the seabed.

The magnetic permeability of the seabed can be taken to be the same as the permeability of free space.

There is a current of 1.80 kA in each cable.

The cables are buried 30.0 m apart, as shown in Figure 15A.

not to scale



Exam question — 2022 Q15(a) (2 of 2)

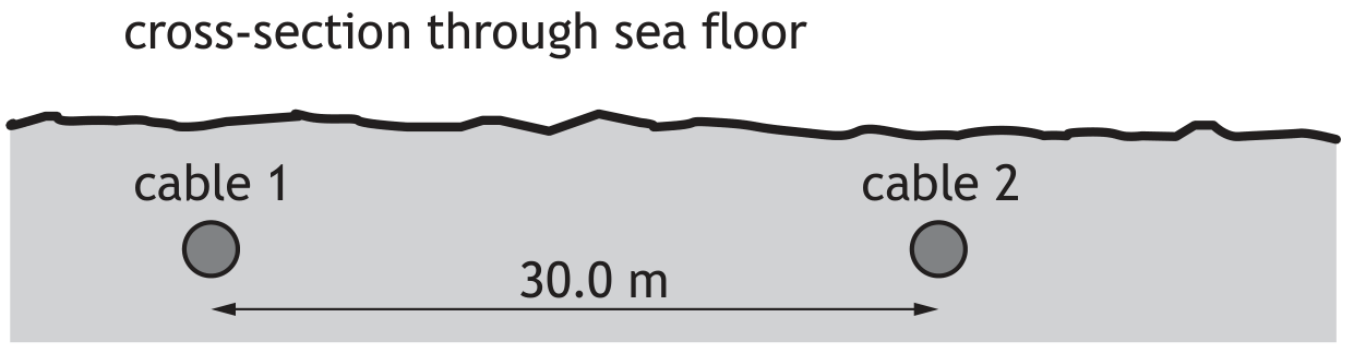


Figure 15A

(a) (i) Calculate the magnetic induction at cable 2 due to the current in cable 1. **3**

Space for working and answer

(ii) Determine the magnitude of the **force per unit length** acting on cable 2 due to the current in cable 1. **3**

Space for working and answer

Marking — 2022 Q15(a)

Question			Expected response	Max mark	Additional guidance
15.	(a)	(i)	$B = \frac{\mu_0 I}{2\pi r}$ (1) $B = \frac{4\pi \times 10^{-7} \times 1.8 \times 10^{3}}{2\pi \times 30.0}$ (1) $B = 1.2 \times 10^{-5}$ T (1)	3	Accept: 1, 1.20, 1.200
		(ii)	$F = IlB$ (1) $\frac{F}{l} = 1.8 \times 10^{3} \times 1.2 \times 10^{-5}$ (1) $\frac{F}{l} = 2.2 \times 10^{-2}$ Nm^{-1} (1)	3	or consistent with (a)(i) Accept: 2, 2.16, 2.160 Where l is substituted as 1 accept final answer in N

Exam question — 2026 Q15(a)(iii)

15. A model electric motor consists of a rotor, which is free to rotate about an axis perpendicular to a uniform magnetic field, as shown in **Figure 15A**.

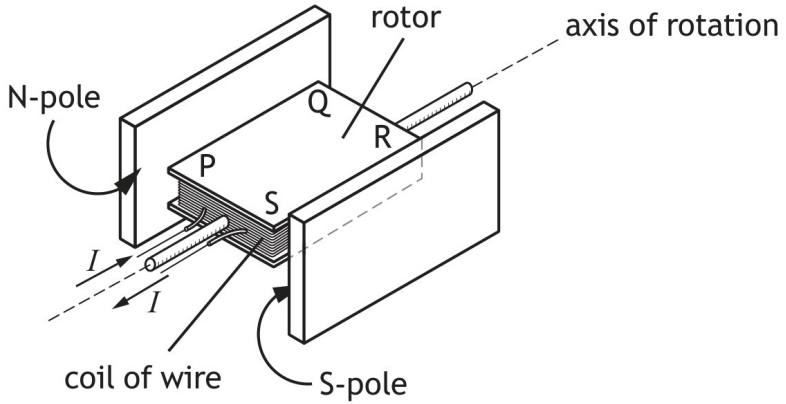


Figure 15A

The length of the wire along the side PQ is 45 mm.

The magnetic induction between the poles of the magnets is 44 mT.

The current in the coil of wire is 0.56 A.

(a) (i) Calculate the magnitude of the force experienced by one wire of the coil on the rotor side PQ. **3**

Space for working and answer

(iii) Explain why no force acts on the rotor side QR. **1**

[Turn over

Marking — 2026 Q15(a)(iii)

Question			Expected response	Max mark	Additional guidance
		(iii)	Current direction is parallel to the magnetic field.	1	

Exam question — 2026 Q15(b)(i)

(b) The rotor now rotates 90° clockwise to the position shown in **Figure 15B**.

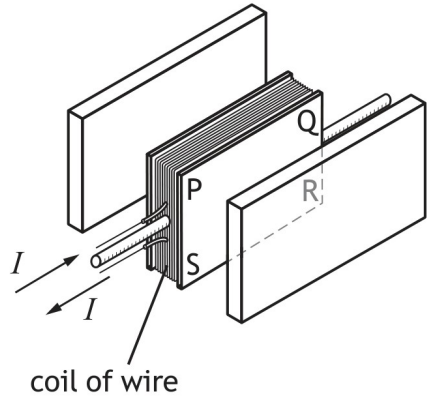


Figure 15B

(i) On **Figure 15C**, show the direction of the forces acting on sides PQ and RS of the rotor. **1**

(An additional diagram, if required, can be found on *page 50*.)

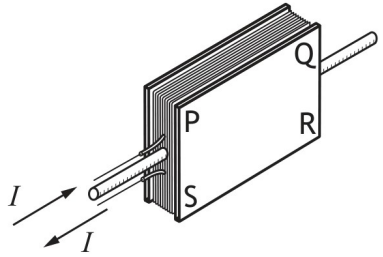
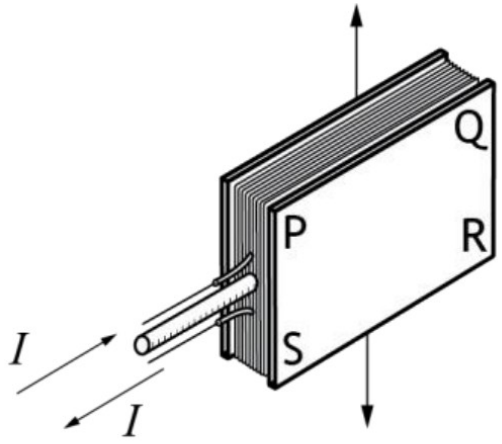


Figure 15C

Marking — 2026 Q15(b)(i)

Question			Expected response	Max mark	Additional guidance
	(b)	(i)		1	Both arrows are required.

Exam question — 2026 Q15(c)

(c) The rotor is modified so that there are now two perpendicular coils of wire, as shown in **Figure 15D**.

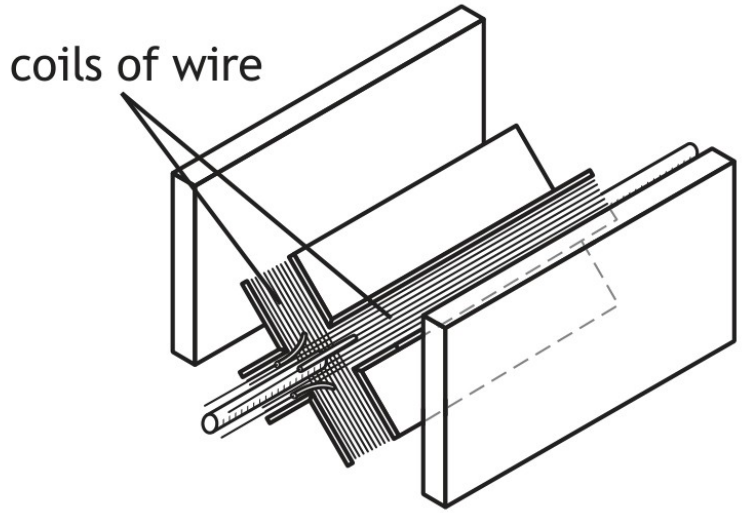


Figure 15D

Explain why this modification improves the operation of the model electric motor.

1

[Turn over

Marking — 2026 Q15(c)

Question			Expected response	Max mark	Additional guidance
	(c)		There is always a (component of) force perpendicular (to the axis of rotation).	1	Torque is never zero (when current in coil).

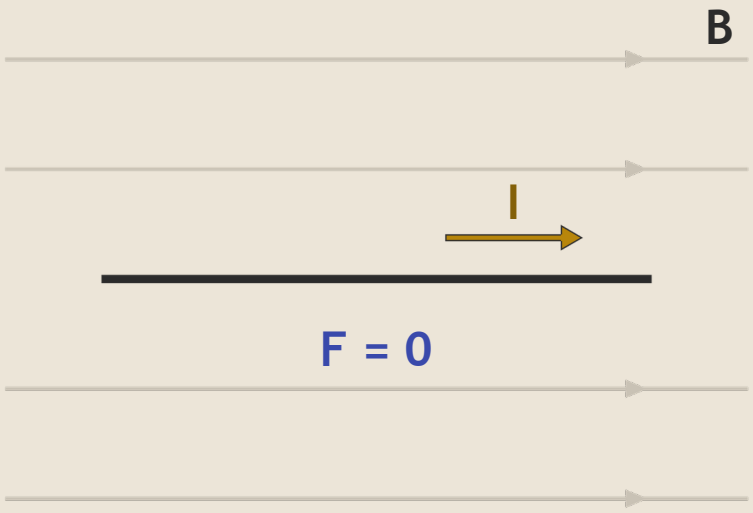
What Candidates Get Wrong

No force on QR: current **parallel** to the field (2026 Q15(a)(iii): few).

Coil forces: draw **both** arrows (2026 Q15(b)(i) MI).

Two coils: always a force **perpendicular** to the axis (2026 Q15(c)).

Per unit length: unit N m^{-1} (2022 Q15(a)(ii) MI).



Look back — by the end of this part...

1. I can use $B = \mu_0 I / 2\pi r$.
2. I can use $F = IlB \sin \theta$.
3. I can **calculate** the force between two wires.
4. I can **explain** when there is **no force**.

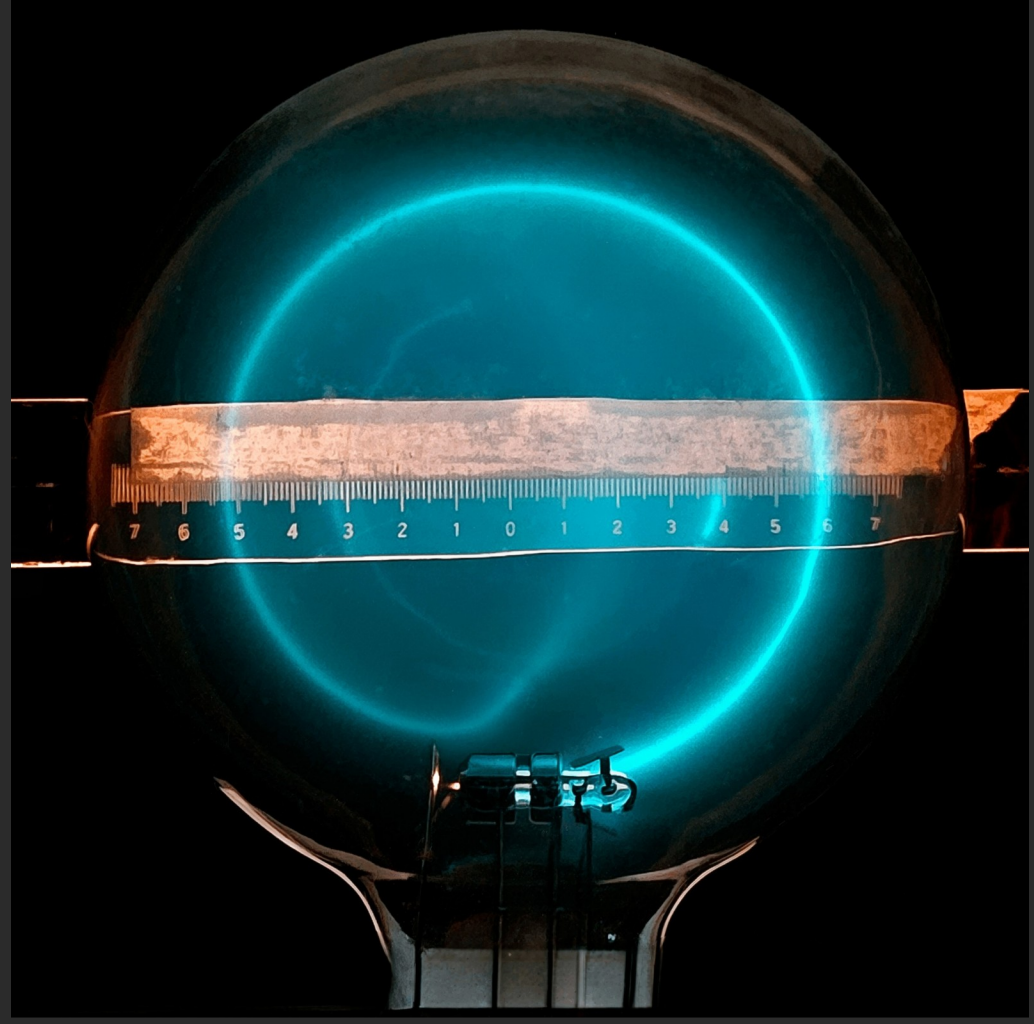
PART 3

Charged particles in a magnetic field

Circles, spirals and a machine that picks one speed.

By the end of this part...

1. I can use $F = qvB$.
2. I can **show** that $r = mv/qB$.
3. I can **explain** the **helical** path.
4. I can use $qE = qvB$ in a **velocity selector**.



An electron beam bent into a circle. What keeps it going round?

Force on a moving charge

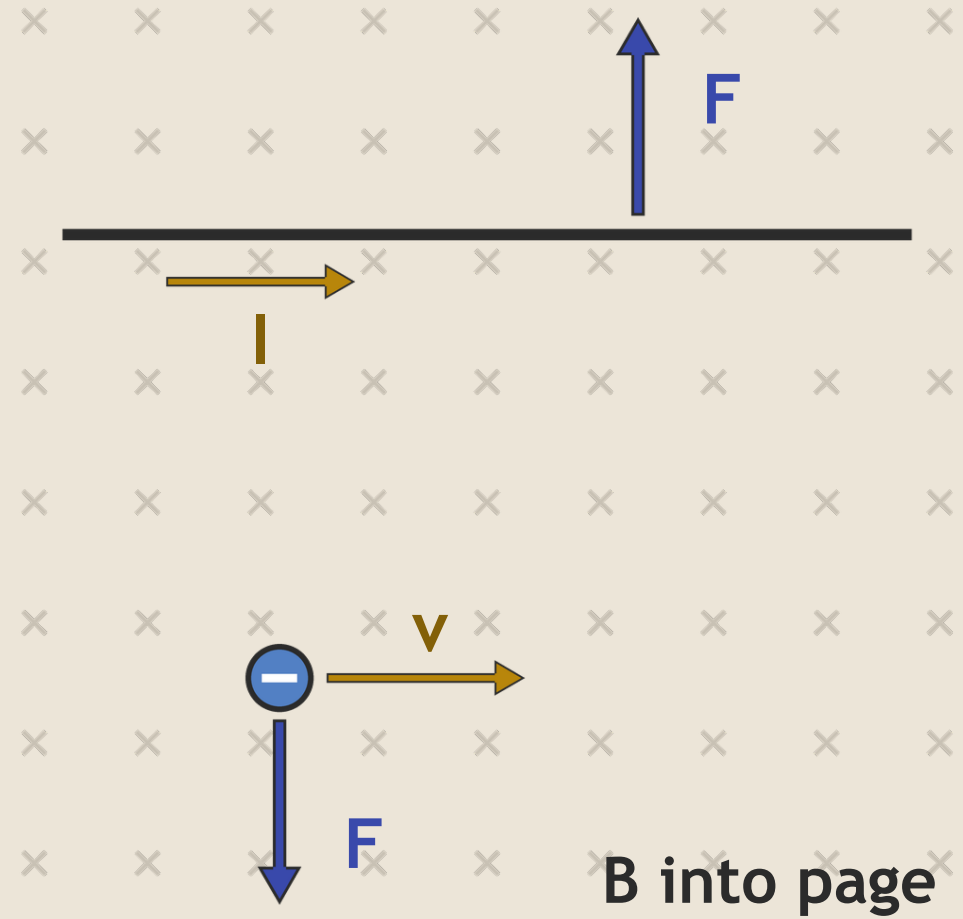
$$F = qvB$$

q: charge (C)

v: speed (m s^{-1}), at 90° to B

B: magnetic induction (T)

Same idea as F = IlB.



Into the page, out of the page

×

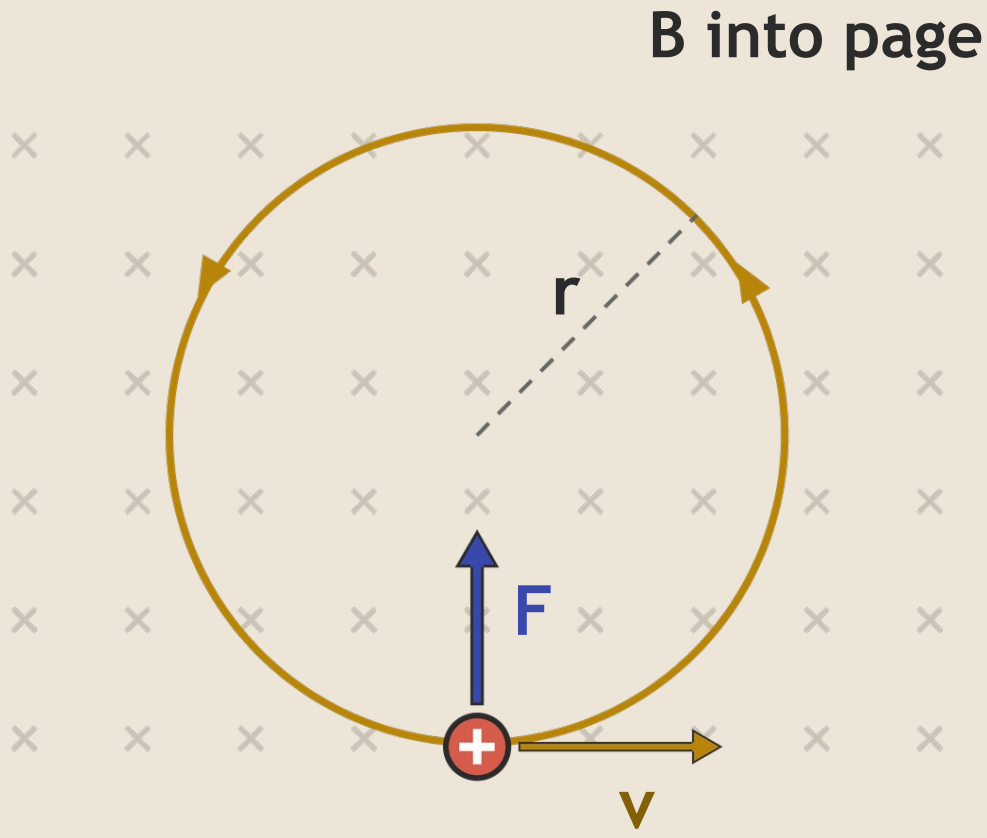
B **into** the page.

•

B **out of** the page.

In the exam

Write "into the page". Never "down".



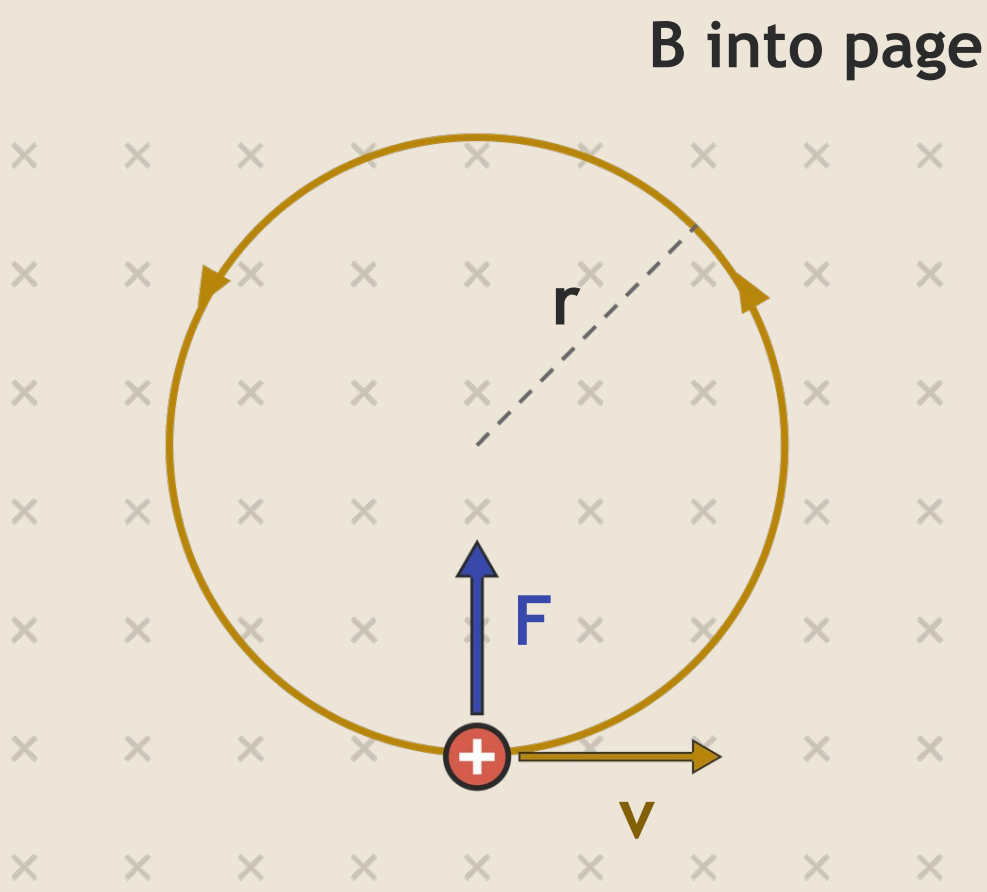
Why the path is a circle

$$qvB = mv^2/r$$

$$r = mv/qB$$

F always at 90° to v,

so the speed never changes.



Exam question — 2021 Q14(b)(i)

(b) (i) By considering the force acting on a proton as it moves in a semi-circular path, show that its speed at point Y is

$$v = \frac{qBr}{m}$$

where the symbols have their usual meaning. **2**

How it's marked — 2021 Q14(b)(i)

Question			Expected response	Max mark	Additional guidance
	(b)	(i)	$(F = qvB)$ $\left(F = \frac{mv^2}{r}\right)$ $\frac{mv^2}{r} = qvB$ (1),(1) $v = \frac{qBr}{m}$	2	**SHOW** 1 for both relationships 1 for equating Alternative method $(F = qvB)$ $\left(F = mr\omega^2\right)$ $(v = r\omega)$ $mr\left(\frac{v}{r}\right)^2 = qvB$ $v = \frac{qBr}{m}$ 1 mark for **all** relationships 1 mark for substitution for ω and equating Final relationship must be shown or max 1.

Worked example 4 — radius of the path

A proton moves at **2.0×10^{6} m s^{-1}** at right angles to a magnetic field of **0.50 T**.

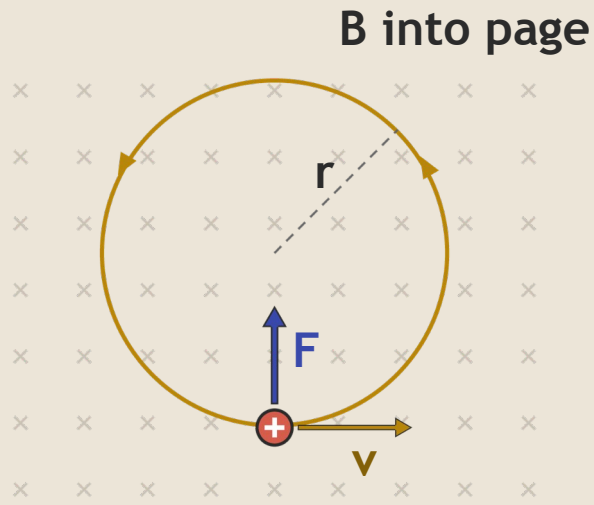
Calculate the radius of its path. **3**

$$qvB = mv^2/r, \text{ so } r = mv/qB$$

$$r = 1.673 \times 10^{-27} \times 2.0 \times 10^{6} / (1.60 \times 10^{-19} \times 0.50)$$

$$r = 4.2 \times 10^{-2} \text{ m}$$

3 marks: relationship, substitution, answer with unit.



Moving at an angle to B — a helix

$v_{\parallel}$ along B

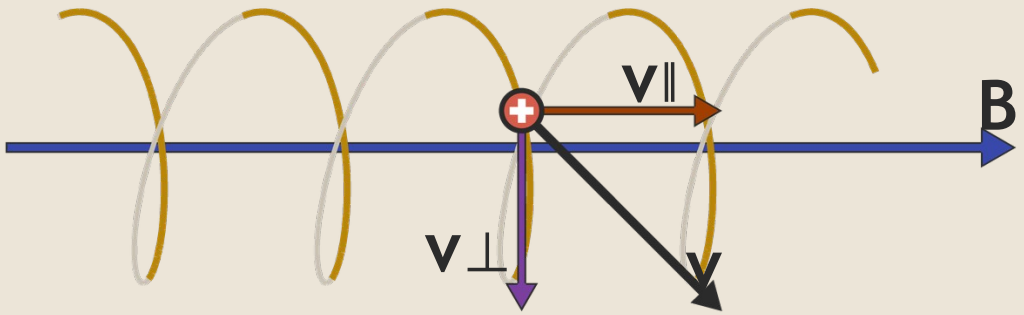
No force: stays the same.

$v_{\perp}$ across B

Force $qv_{\perp}B$: a circle.

Together

A **helix** round the field line.



Explaining the helix

1. **Split** the velocity: $v_{\parallel}$ along B and $v_{\perp}$ across B.
2. $v_{\parallel}$ feels **no force**, so the particle moves along B at a steady speed.
3. $v_{\perp}$ feels a force at 90° to it, so the particle goes round in a **circle**.
4. Both at once: a **helix** along the field line.

Exam question — 2021 Q14(b)(ii)

(b) (i) By considering the force acting on a proton as it moves in a semi-circular path, show that its speed at point Y is

$$v = \frac{qBr}{m}$$

where the symbols have their usual meaning. **2**

(ii) The magnetic induction is 0·714 T. The radius of the semi-circular path at Y is 0·105 m.

Calculate the speed of a proton at point Y. **2**

Space for working and answer

Marking — 2021 Q14(b)(ii)

Question			Expected response	Max mark	Additional guidance
		(ii)	$v = \frac{qBr}{m}$ $v = \frac{1{\cdot}60 \times 10^{-19} \times 0{\cdot}714 \times 0{\cdot}105}{1{\cdot}673 \times 10^{-27}}$ (1) $v = 7{\cdot}17 \times 10^{6}\ \text{ms}^{-1}$ (1)	2	Accept: 7·2, 7·170, 7·1700

Exam question — 2021 Q14(a) (1 of 2)

14. Proton beam therapy is a medical treatment. Protons are accelerated to specific velocities using a cyclotron.

A cyclotron is a particle accelerator that consists of two D-shaped hollow structures, called Dees, placed in a vacuum. The Dees are separated by a gap.

This is shown in **Figure 14A**.

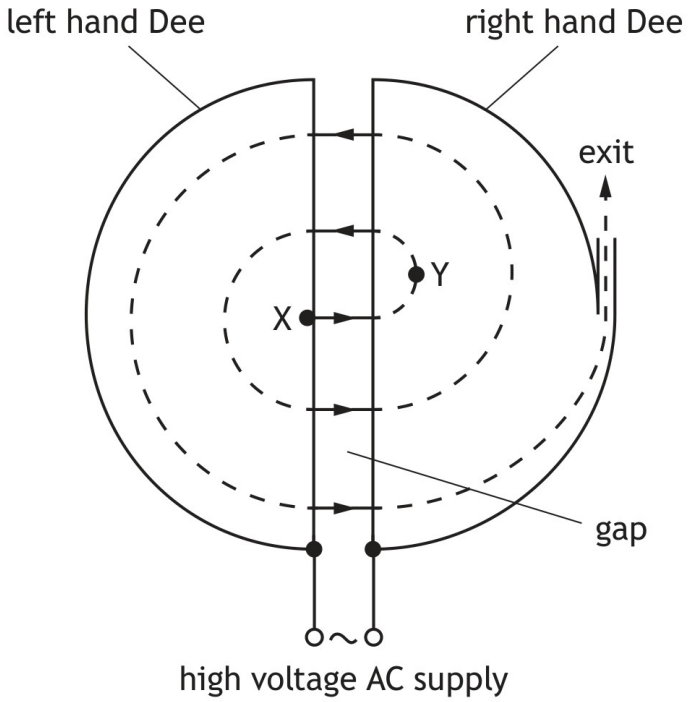


Figure 14A

Exam question — 2021 Q14(a) (2 of 2)

During testing, protons are introduced to the cyclotron at point X.

The protons are accelerated from rest across the gap by an electric field.

Inside the Dees there is a uniform magnetic field B.

This field acts on the protons causing them to move in semi-circular paths within the Dees.

(a) Determine the direction of the magnetic field B. **1**

Marking — 2021 Q14(a)

Question			Expected response	Max mark	Additional guidance
14.	(a)		Into the page.	1	Do not accept ‘down’.

Exam question — 2021 Q14(c)

(c) Explain why an AC supply must be used to provide the electric field. **1**

Marking — 2021 Q14(c)

Question			Expected response	Max mark	Additional guidance
	(c)		The direction of (electrical) force acting on the proton must change (every time the proton crosses the gap).	1	Accept The proton travels in opposite directions (every time the proton crosses the gap).

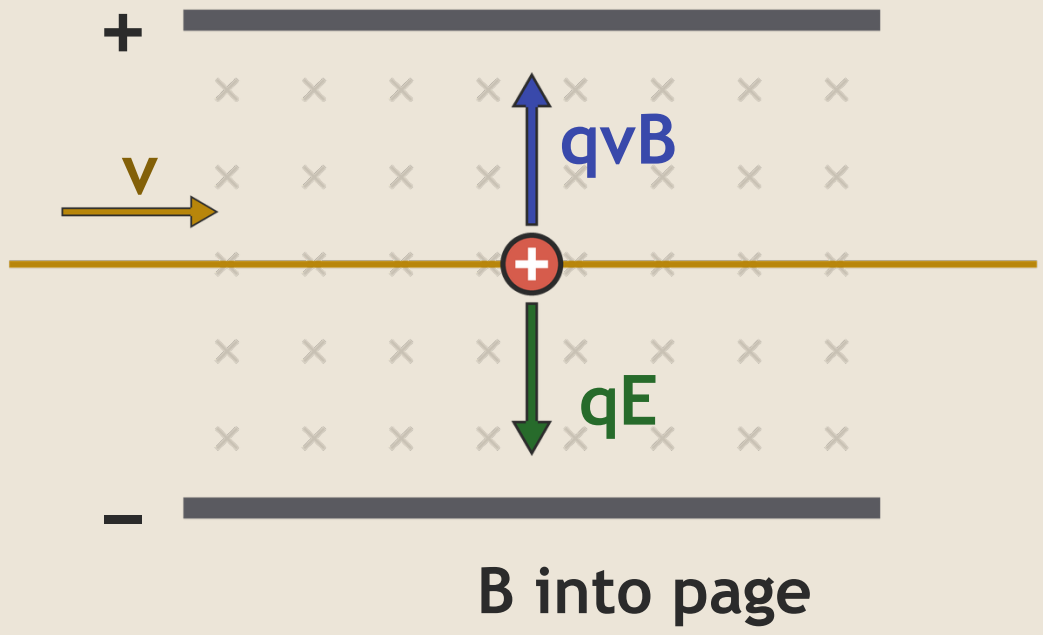
Velocity selector

$$qE = qvB$$

$$v = E/B$$

E and B at 90° to each other.

Only one speed goes straight.



Exam question — 2025 Q14(a)(iii)

(a) (i) By considering the electric and magnetic forces acting on an ion, show that the velocity v of an ion passing undeflected through the velocity selector is given by the relationship

$$v = \frac{E}{B}$$

where the symbols have their usual meanings. **2**

Space for working and answer

(iii) Explain why different types of ion passing through the velocity selector undeflected have the same velocity. **1**

How it's marked — 2025 Q14(a)(iii)

Question			Expected response	Max mark	Additional guidance
		(iii)	Velocity is dependent on E and B only.	1	Velocity is independent of mass and charge.

Worked example 5 — selector, then a semicircle

In a velocity selector the plates are **20 mm** apart with **2.0 kV** across them. B = **0.25 T**. Protons leave it and enter a field of **0.40 T**.

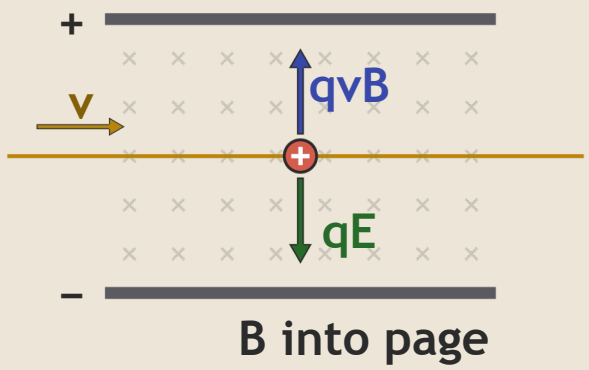
(a) **Determine** the speed of the protons. **3**

$E = V/d = 2.0 \times 10^{3} \ / \ 0.020 = 1.0 \times 10^{5}$

$qE = qvB$, so $v = E/B = 1.0 \times 10^{5} \ / \ 0.25$

$v = 4.0 \times 10^{5}$ m s^{-1}

3 marks: relationships, substitution, answer with unit.



Worked example 5 — continued

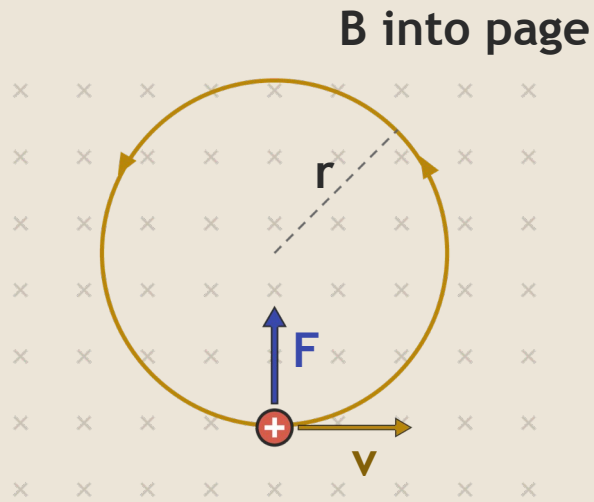
The protons leave the selector at **4.0 × 10^5 m s^{-1}** and enter a field of **0.40 T** at right angles.

(b) **Calculate** the radius of their path in the 0.40 T field. **3**

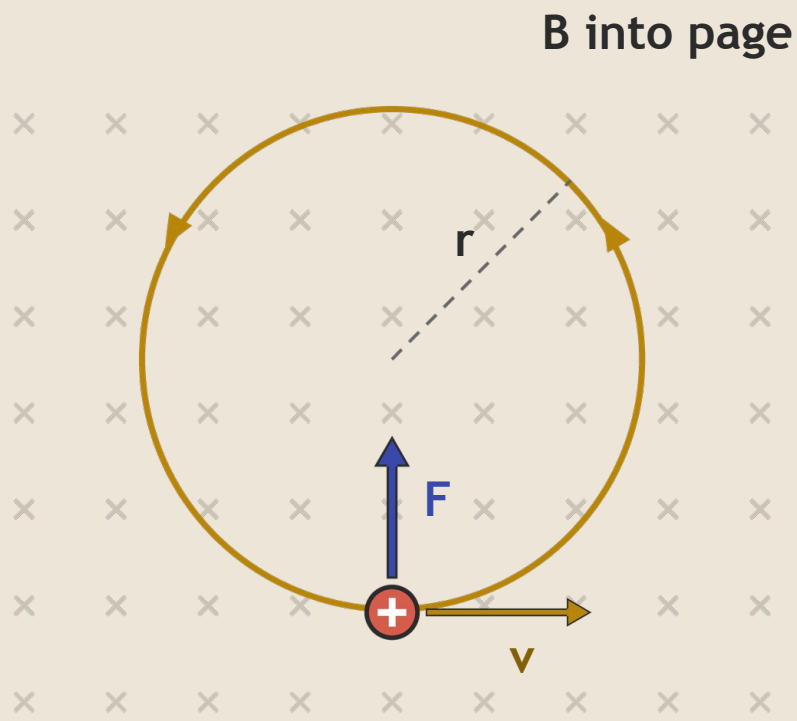
$$r = mv/qB = 1.673 \times 10^{-27} \times 4.0 \times 10^5 / (1.60 \times 10^{-19} \times 0.40)$$

$$r = 1.0 \times 10^{-2} \text{ m}$$

3 marks: relationship, substitution, answer with unit.



An electron moves at **3.0 × 10^{7} m s^{-1}** at right angles to a field of **2.0 mT**. **Calculate** the radius of its path.



r = 8.5 × 10^{-2} m

Exam question — 2025 Q14(a)(ii)

The electric field is produced by a potential difference of 4.80 kV between two parallel plates. The plates are separated by a distance of 10.0 mm.

This is shown in **Figure 14B**.

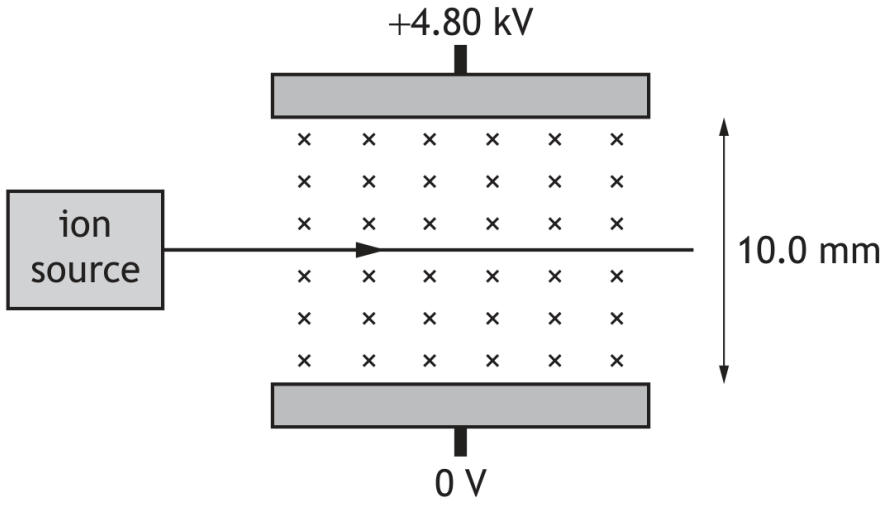


Figure 14B

The magnetic induction in the velocity selector is 225 mT.

(ii) Determine the velocity of an ion that passes through the velocity selector undeflected. **3**

Space for working and answer

Marking — 2025 Q14(a)(ii)

Question			Expected response	Max mark	Additional guidance
		(ii)	$E = \frac{V}{d}$ (1) $\left(v = \frac{E}{B}\right)$ $v = \frac{\left(\frac{4.80\times10^{3}}{10.0\times10^{-3}}\right)}{225\times10^{-3}}$ (1) $v = 2.13\times10^{6}\ \text{ms}^{-1}$ (1)	3	Accept 2.1, 2.133, 2.1333

Exam question — 2025 Q14(b) (1 of 2)

(b) The mass spectrometer is now used to investigate the paths followed by two types of hydrogen ion.

Data for the hydrogen ions are shown below.

Ion	Symbol	Mass ($\times 10^{-27}$ kg)	Charge ($\times 10^{-19}$ C)
Deuterium	$^{2}H^{+}$	3.34	1.60
Tritium	$^{3}H^{+}$	5.01	1.60

These ions enter the deflection field with the velocity determined in (a) (ii) and follow semi-circular paths in the field.

One type of ion leaves the field at point Y and the other type of ion leaves at point Z.

This is shown in **Figure 14C**.

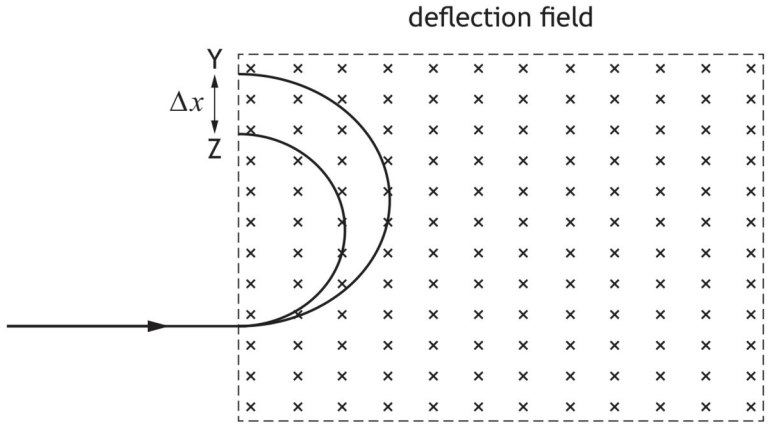


Figure 14C

Exam question — 2025 Q14(b) (2 of 2)

The magnetic induction in the deflection field is 0.65 T.

The radius of the curved path of the ions is given by the relationship

$$r = \frac{mv}{qB}$$

where the symbols have their usual meanings.

Determine the separation Δx of the ions when they leave the deflection field. **4**

Space for working and answer

Marking — 2025 Q14(b)

Question			Expected response	Max mark	Additional guidance
	(b)		$r = \frac{mv}{qB}$ $r_{2_H} = \frac{3.34\times10^{-27}\times2.13\times10^{6}}{1.60\times10^{-19}\times0.65}$ (1) $r_{3_H} = \frac{5.01\times10^{-27}\times2.13\times10^{6}}{1.60\times10^{-19}\times0.65}$ (1) $\Delta x = 2\times\left(\left(\frac{5.01\times10^{-27}\times2.13\times10^{6}}{1.60\times10^{-19}\times0.65}\right)-\left(\frac{3.34\times10^{-27}\times2.13\times10^{6}}{1.60\times10^{-19}\times0.65}\right)\right)$ (1) $\Delta x = 0.068$ m (1)	4	Accept 0.07, 0.0684, 0.06841 Or consistent with (a)(ii)

Exam question — 2025 Q14(c)

(c) The potential difference between the parallel plates in the velocity selector is now increased. The magnetic inductions of both the velocity selector and the deflection field are unchanged.

State the change to the paths of the ions in the deflection field.

Justify your answer. **2**

Marking — 2025 Q14(c)

Question			Expected response		Max mark	Additional guidance
	(c)		Radii (of curvature) increase	(1)	**2**	Justify
			Velocity of ions increases	(1)		

Exam question — 2019 Q15(b)(i) (1 of 2)

(b) After a short time, the bottom of the plate becomes negatively charged relative to the top of the plate, as shown in Figure 15C.

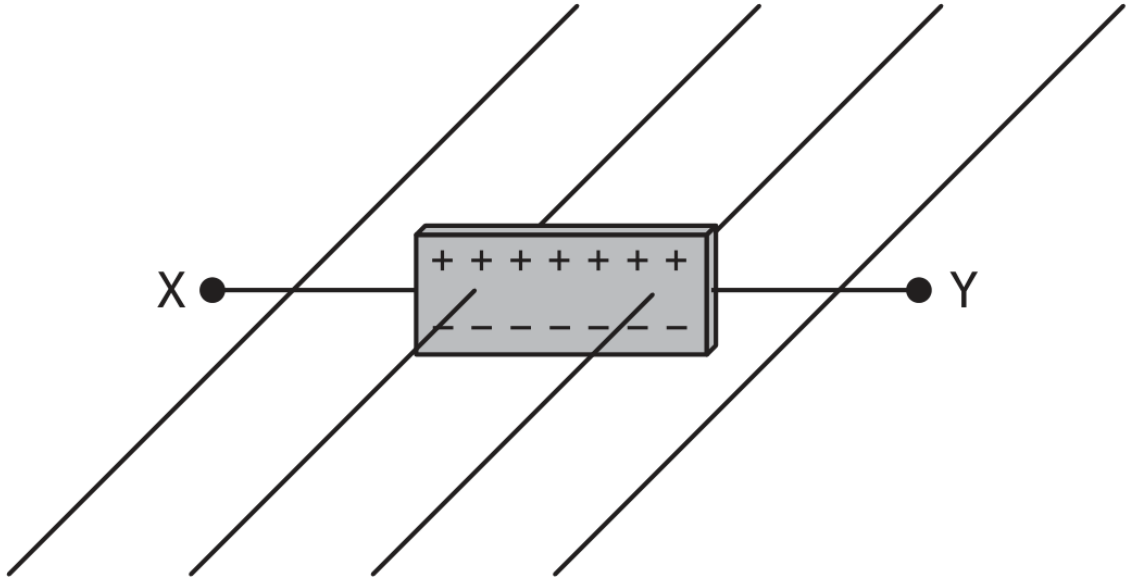


Figure 15C

This causes a uniform electric field between the top and bottom of the metal plate.

Exam question — 2019 Q15(b)(i) (2 of 2)

Electrons moving at a fixed speed v_d, called the *drift velocity*, will now travel horizontally across the plate. These electrons do not move vertically as the electric and magnetic forces acting on them are balanced.

(i) Show that the drift velocity is given by the relationship

$$v_d = \frac{V}{Bd}$$

where

V is the potential difference between the top and bottom of the metal plate

B is the magnetic induction

d is the height of the metal plate. **3**

Marking — 2019 Q15(b)(i)

Question			Expected response	Max mark	Additional guidance
	(b)	(i)	$\left(F_{(E)} = QE, F_{(B)} = qvB\right)$ $EQ = qvB$ **(1) , (1)** $E = v\ B$ $V = Ed$ **(1)** $v = \frac{V}{Bd}$	**3**	SHOW QUESTION **(1 mark)** for both relationships, **(1 mark)** for equality of forces or fields Final line must appear or max **(2 marks)**

Exam question — 2019 Q15(b)(iii)

(b) After a short time, the bottom of the plate becomes negatively charged relative to the top of the plate, as shown in Figure 15C.

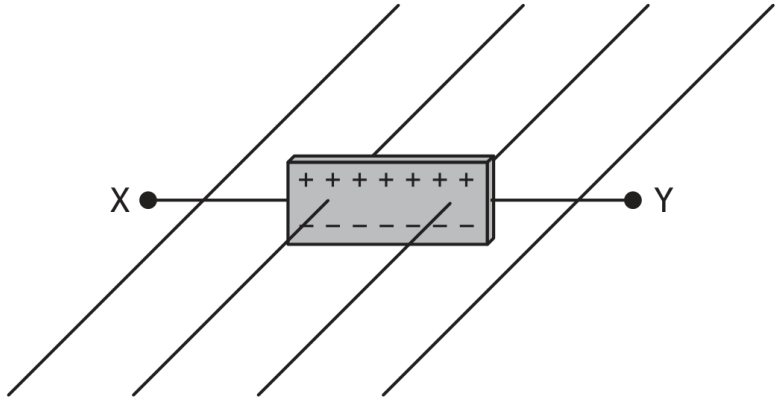


Figure 15C

This causes a uniform electric field between the top and bottom of the metal plate.

Electrons moving at a fixed speed v_d, called the *drift velocity*, will now travel horizontally across the plate. These electrons do not move vertically as the electric and magnetic forces acting on them are balanced.

(iii) The magnetic induction is now increased. The drift velocity of the electrons moving through the metal plate remains the same.

Explain why the drift velocity does not change. **2**

Marking — 2019 Q15(b)(iii)

Question			Expected response	Max mark	Additional guidance
		(iii)	Because more charges have been separated (vertically) across the plate. **OR** More electrons gather on the bottom of the plate. **(1)** (The increased magnetic force) increases the electric force/potential difference/electric field strength (across the plate). **(1)**	**2**	Marks are independent.

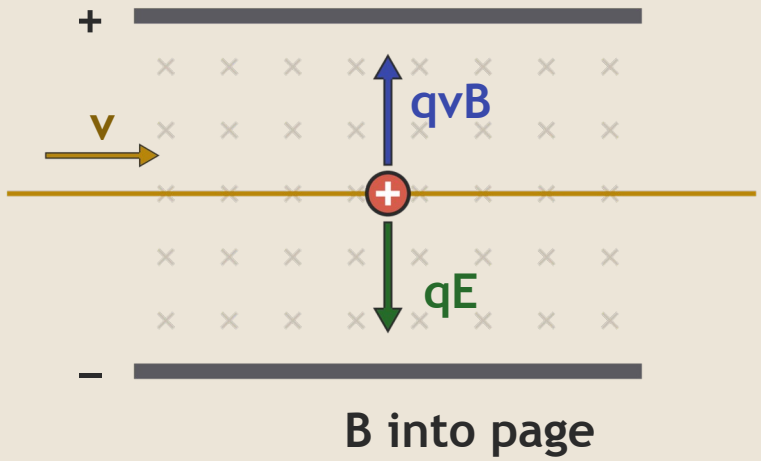
What Candidates Get Wrong

Direction: "into the page", never "down" (2021 Q14(a) MI).

Show: write the last line, $v = qBr/m$, or 1 mark at most (2021 Q14(b)(i) MI).

Selector: v depends on **E and B only** (2025 Q14(a)(iii): only some).

Two ions: both radii, then $\Delta x = 2(r_2 - r_1)$ (2025 Q14(b): few).



Look back — by the end of this part...



1. I can use $F = qvB$.
2. I can **show** that $r = mv/qB$.
3. I can **explain** the **helical** path.
4. I can use $qE = qvB$ in a **velocity selector**.

Magnetic fields in the exam

Year	Question	Marks
2026	Q15	11
2025	Q14	12
2024	Q14	7
2023	Q10 (c), Q14	14
2022	Q15 (a)	6

Year	Question	Marks
2021	Q14	6
2019	Q15	8
2018	Q14	11
2016	Q14	10

Past paper finder

Reading and homework

Scholar Magnetic fields

Scholar — Unit 3, Magnetic fields, and its end-of-topic test
BrightRed Study Guide pages 98-103
Past papers — the questions on the last slide

Scholar

Errors that cost marks

- Solenoid: straight, parallel lines inside, and arrows.
- Ferromagnetic: dipoles can be made to align.
- No force when the current is parallel to B.
- "Into the page", never "down".
- Velocity selector: v depends on E and B only.

Check yourself

1. **Sketch** the field round a solenoid and round Earth.
2. **State** what ferromagnetism is.
3. **Calculate** B near a wire and the force on a wire.
4. **Show** that r = mv/qB, and **explain** a helix.
5. **Use** qE = qvB in a velocity selector.

Experiments you could try

- The F = BIL lab page: a wire, a magnet and a top-pan balance.
- PhET "Magnets and Electromagnets": move a compass round a coil.

- A compass beside a wire: switch the current on and off.
- Iron filings round a bar magnet, then round two magnets.

Course specification — 1 of 4

- Knowledge that electrons are in motion around atomic nuclei and individually produce a magnetic effect.
- Knowledge that, for example, iron, nickel, cobalt, and some rare earths exhibit a magnetic effect called ferromagnetism, in which magnetic dipoles can be made to align, resulting in the material becoming magnetised.



Course spec, pages 18-19

Course specification — 2 of 4

- Sketch of magnetic field patterns between magnetic poles and around solenoids, including the magnetic field pattern around Earth.
- Comparison of gravitational, electrostatic, magnetic, and nuclear forces in terms of their relative strength and range.



Course spec, pages 18-19

Course specification — 3 of 4

- Use of an appropriate relationship to solve problems involving magnetic induction around a current-carrying wire, the current in the wire and the distance from the wire. $B = \mu_0 I / 2\pi r$



Course spec, pages 18-19

Course specification — 4 of 4

- Explanation of the helical path followed by a moving charged particle in a magnetic field.
- Use of appropriate relationships to solve problems involving the forces acting on a current-carrying wire in a magnetic field and a charged particle in a magnetic field.
- $F = IlB \sin \theta$ $F = qvB$ $F = mv^2 / r$

Course spec, pages 18-19

Credits (1 of 2)

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Credits (2 of 2)

Magnetic fields in the exam: © Qualifications Scotland (SQA). AH Physics 2016-2026.

Reading and homework: SCHOLAR AH Physics © Heriot-Watt University. BrightRed AH Physics Study Guide

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R Stewart - November 2026